Team 1 - Lab 1

Forecasting with ARIMA and ETS models

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April 21, 2025

Contents

Data

This lab used annual data recording salmon landings in six different US states covering the period from 1950 to 2016: Alaska, California, Oregon, Michigan, Pennsylvania, and Washington. Due to sparse data, Pennsylvania and Michigan were excluded from the analysis. Monthly data recording salmon landings as well as the value of fish landed was used from Washington, Oregon, and California covering the period from 1990 to 2022.

Research Questions

1. What model structure best fits the different states and regions (Total and Pacific Northwest) represented in the yearly data set? How does forecasting accuracy compare across these regions?
2. How does model performance change when sections of the time series are dropped?
3. How does forecasting monthly value of catch compare with separate forecasts of monthly landings and prices being interacted to find overall value?

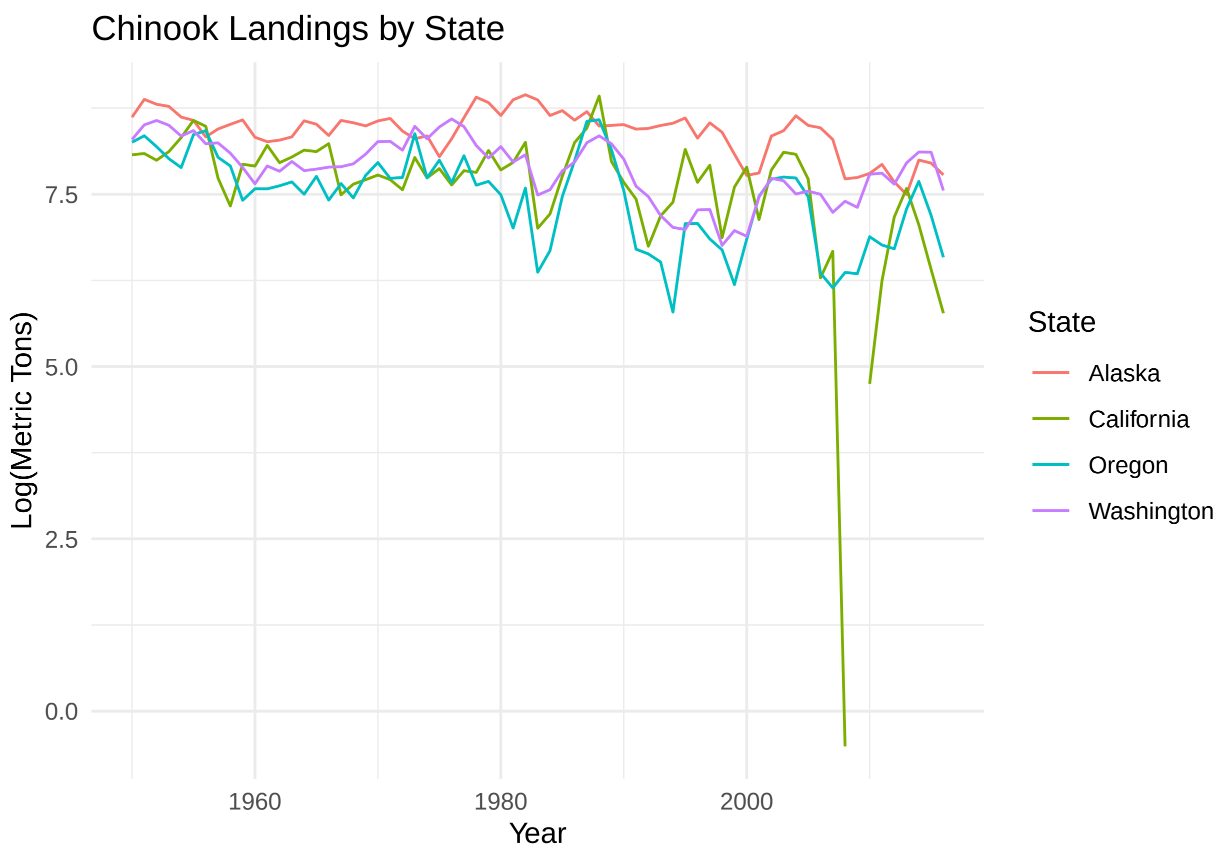
Methods

Initial plan

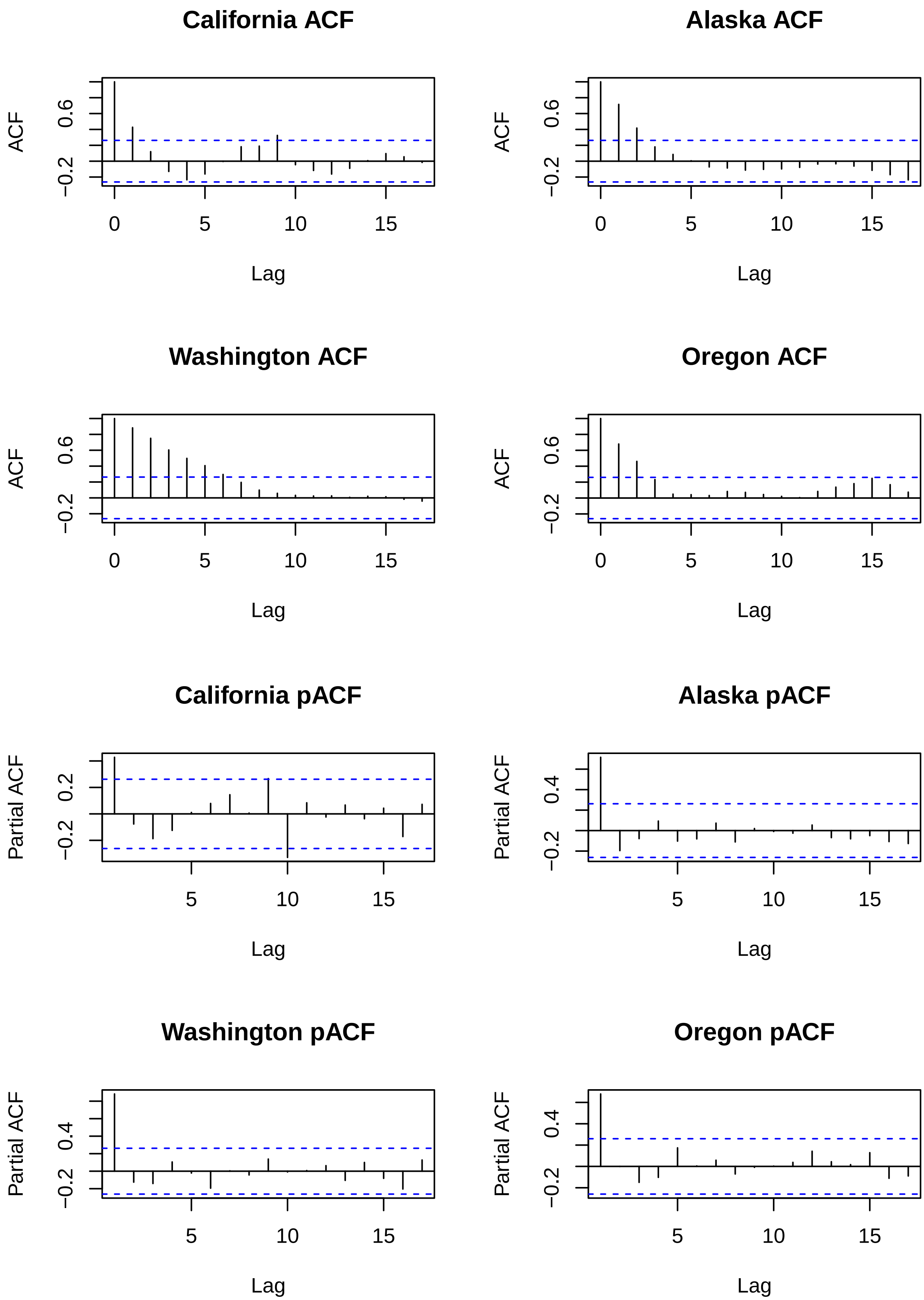
We initially planned to fit ARIMA and ETS models to the different states and compare which model structure best fit each state or region. Noting the similarities between Washington and Oregon's time series data, we also wanted to compare forecast accuracy for models trained on different data sets (e.g. compare Washington's forecast accuracy on Oregon's test data). We also planned on recovering monthly prices by dividing the value of fish landed by the landed weight in order to assess the efficiency of forecasting approaches.

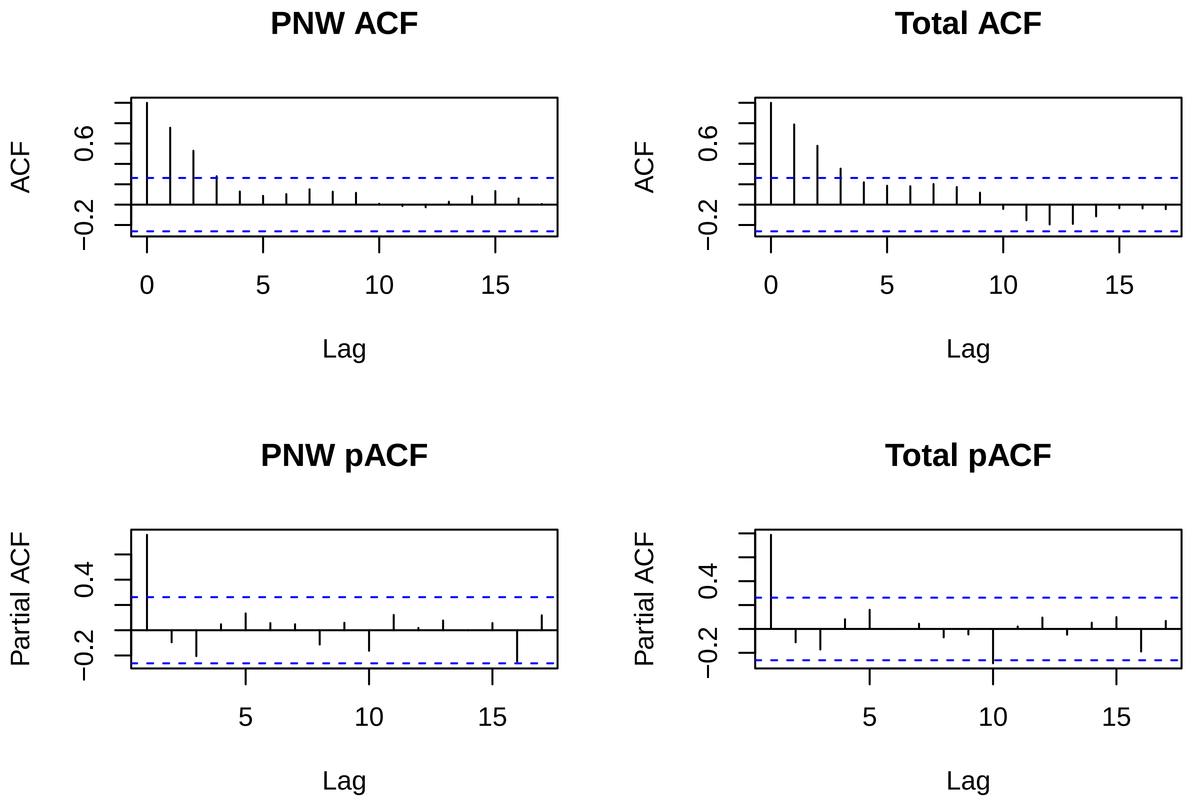
Diagnostics and preliminary exploration - Question 1

To address question 1, we subset the data for each west coast state, Pacific Northwest which includes WA, OR, and CA, and the total of all states combined, into a training data set and test data set. The training data sets spanned the years 1950-2005, and the test data set was 2006-2016. When we plotted the data there were no obvious problems with stationarity. There might be a slight downwards trend, and there are a few gaps in the California data, but it is not immediately obvious if the data is non-stationary.


Chinook Landings by State
Log(Metric Tons)
7.5
5.0
2.5
0.0
1960
1980
2000
Year
State
Alaska
California
Oregon
Washington

Use ACF and PACF





What did you learn? Also try decomposition. *I might need help interpreting these plots*

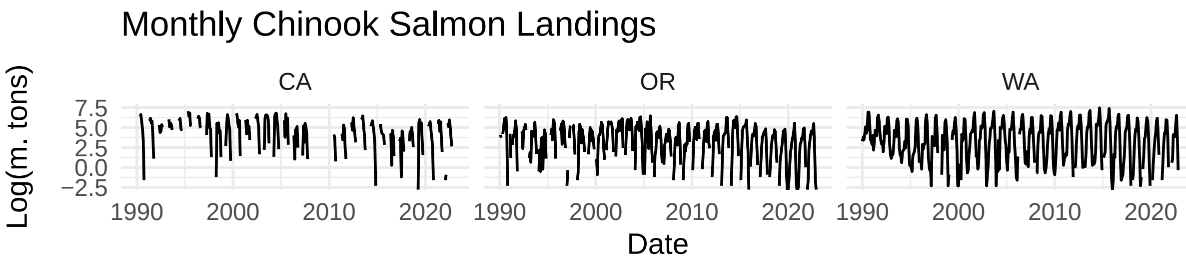
Test for stationarity

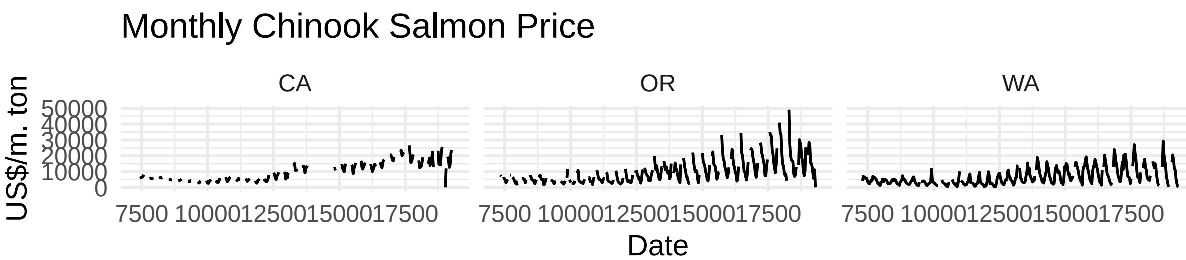
When we run ndiffs on the states' data, Oregon, Washington, the Pacific Northwest region determined one first-difference is required to make the data stationary.

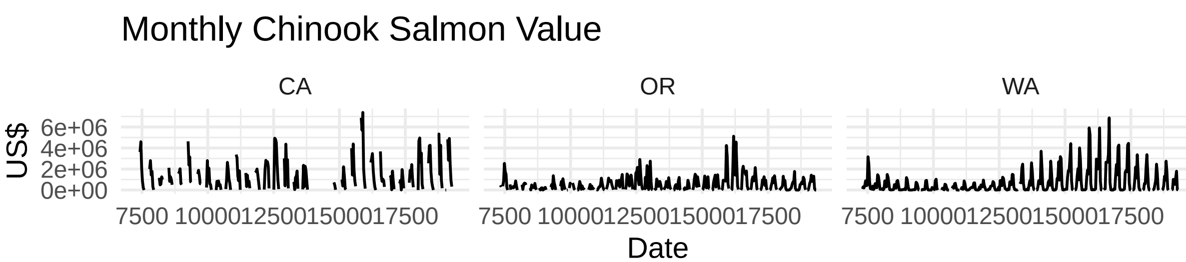
```
##      Alaksa California Oregon Washington Total PNW
## [1,]      0          0      1          1     1   1
```

Diagnostics and preliminary exploration - Question 3

Preliminary analysis of question 2 involved first recovering price data and then plotting monthly landings, prices, and total value for all three states represented.

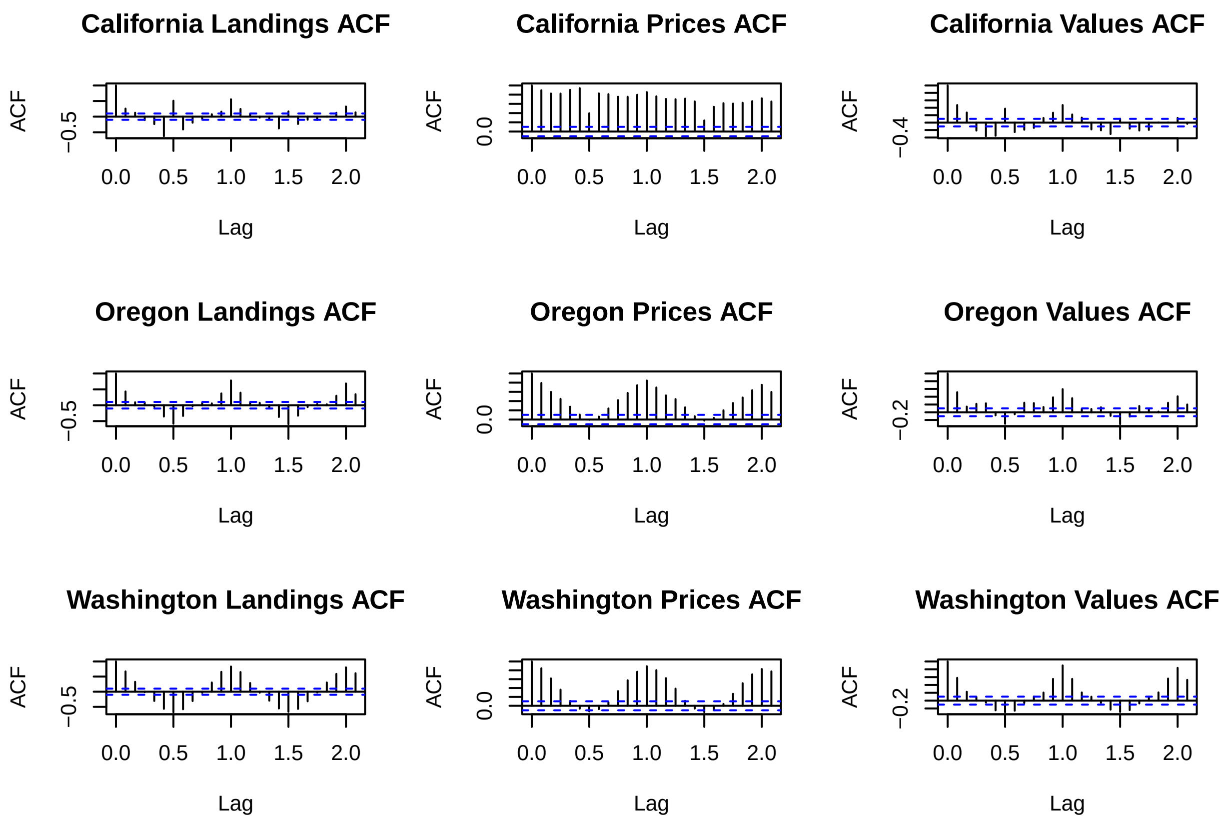


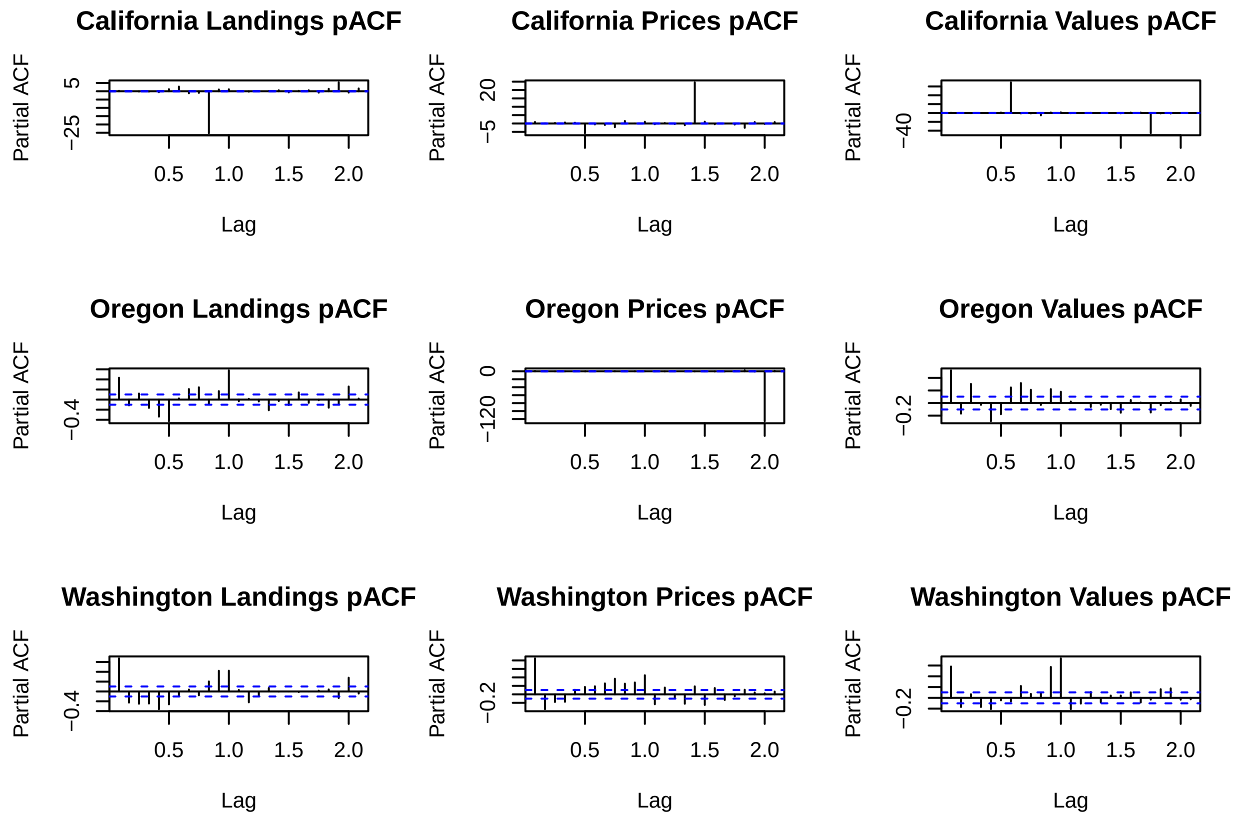




Next, diagnostics were performed on these data, evaluating ACFs and pACFs checking ndiffs for stationarity.

```
##    Landings Prices Value
## CA        1      0     0
## OR        1      1     1
## WA        0      1     1
```


California Landings ACF
California Prices ACF
California Values ACF
Oregon Landings ACF
Oregon Prices ACF
Oregon Values ACF
Washington Landings ACF
Washington Prices ACF
Washington Values ACF
ACF
−0.5
0.0
−0.4
−0.2
0.0 0.5 1.0 1.5 2.0
Lag



There is clear evidence of seasonal trends in the above plots, meaning any useful model will more than likely involve a seasonal component (sARIMA).

Finally, data from each of three state-level time series was subsetted into training and testing data sets. Training data spanned the years from 1990 through 2020 while test data covers the years 2021 & 2022.

Results

Question 1

We fit multiple ARIMA and Holt models to the yearly time series data for each state, the PNW region (WA, OR, CA), and the total chinook landings for all states combined. ARIMA models where p, d, or q were greater than 1 had significantly higher AIC values than more parsimonious ARIMA models and are excluded from this write-up. We then forecasted 10 years of data using the six ARIMA models, tested the forecasts against the test data for each region. Using root mean squared error, we selected the best fit model for each state or region. The Holt model was the best fit for Alaska, California, and the Pacific Northwest region. An ARIMA(1,0,0) model was the best fit for Washington and Oregon. An ARIMA(0,1,0) model was the best fit for the entire Pacific Coast chinook landing data (log metric tons). See the figures below for the six regional time series and forecasts from the best fit model for each. We will also provide a brief interpretation of the best fit models, and check of the residuals using a Ljung-Box test.

```
## Series: aktrain
## ARIMA(1,0,0) with non-zero mean
##
## Coefficients:
```

```
##          ar1    mean
##       0.7069  8.5064
## s.e.  0.0909  0.0716
##
## sigma^2 = 0.02773:  log likelihood = 21.6
## AIC=-37.2   AICc=-36.74   BIC=-31.12
```

```
## Series: catrain
## ARIMA(1,0,0) with non-zero mean
##
## Coefficients:
##          ar1    mean
##       0.4235  7.8462
## s.e.  0.1195  0.0850
##
## sigma^2 = 0.1432:  log likelihood = -24.12
## AIC=54.25   AICc=54.71   BIC=60.32
```

```
## Series: ortrain
## ARIMA(0,1,0)
##
## sigma^2 = 0.2139:  log likelihood = -36.27
## AIC=74.55   AICc=74.62   BIC=76.57
```

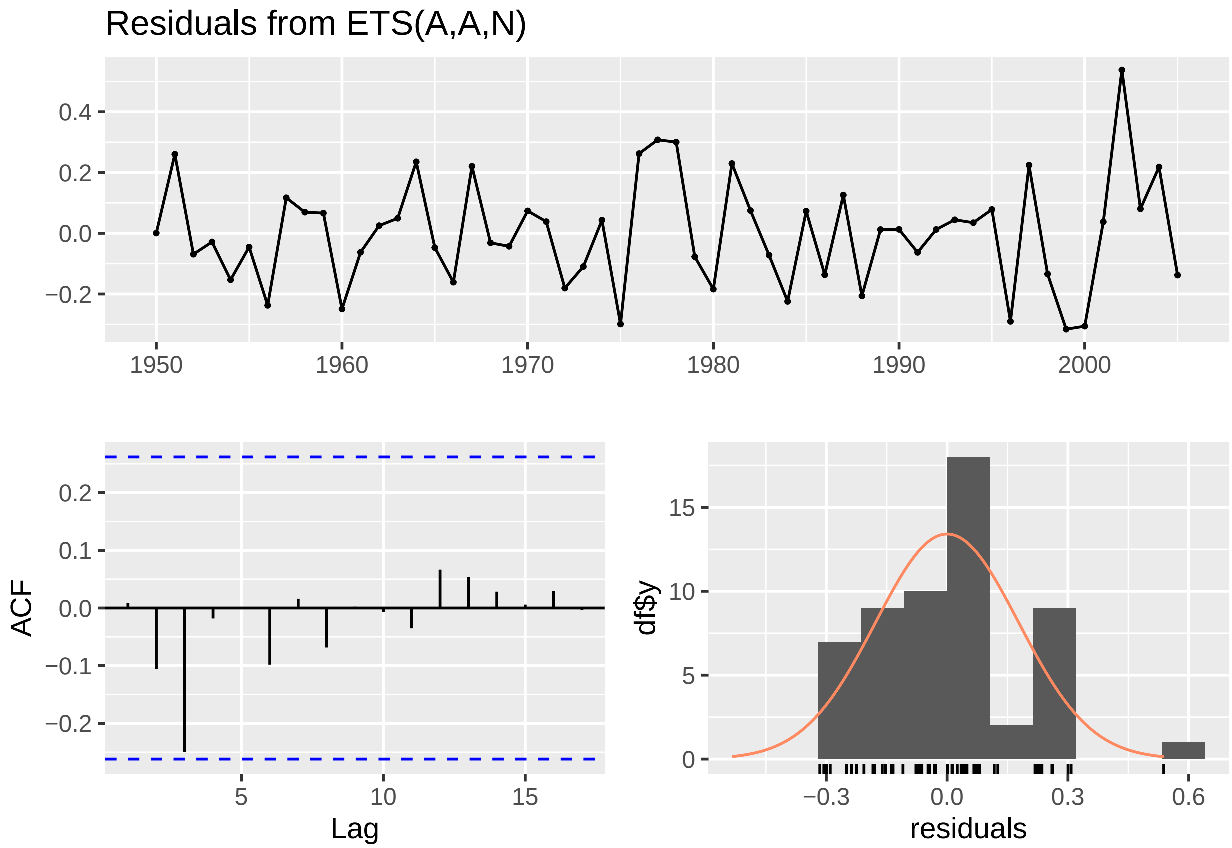
```
## Series: watrain
## ARIMA(0,1,0)
##
## sigma^2 = 0.04634:  log likelihood = 6.43
## AIC=-10.87   AICc=-10.79   BIC=-8.86
```

```
## Series: pnwtrain
## ARIMA(0,1,0)
##
## sigma^2 = 0.08109:  log likelihood = -8.96
## AIC=19.91   AICc=19.99   BIC=21.92
```

```
## Series: totaltrain
## ARIMA(0,1,0)
##
## sigma^2 = 0.03084:  log likelihood = 17.63
## AIC=-33.25   AICc=-33.18   BIC=-31.25
```

Alaska The best fit model for the Alaska chinook landing data was a Holt model with the following equation: $y_t = l_{t-1} + b_{t-1} + \epsilon_t$
$l_t = l_{t-1} + b_{t-1} + 0.9999 * \epsilon_t$
$b_t = b_{t-1} + 0.0001 * \epsilon_t$ This model indicates a slight downward trend that remains fairly consistent over time (beta=0.0001), and gives almost all weight to the most recent observations (alpha=0.9999).

```
##          0,0,0     0,1,0     1,0,0     1,1,0    1,1,1      Holt
## RMSE 0.6611034 0.6579779 0.6641441 0.6567943 0.655056 0.6447745
```

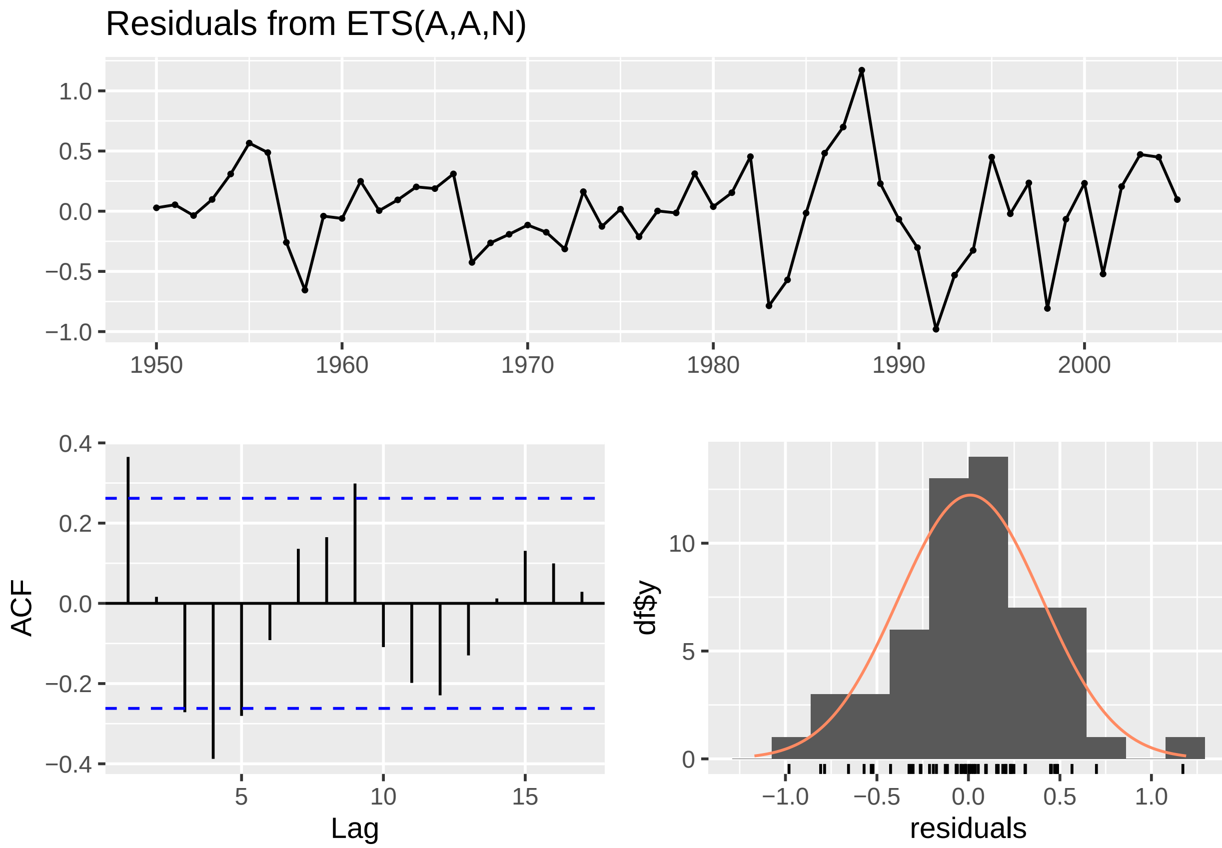



```
##
##  Ljung-Box test
##
## data:  Residuals from ETS(A,A,N)
## Q* = 5.4977, df = 10, p-value = 0.8556
##
## Model df: 0.   Total lags used: 10
```

California The best fit model for California was also a Holt model with the following equation: $y_t = l_{t-1} + b_{t-1} + \epsilon_t$ $l_t = l_{t-1} + b_{t-1} + 0.0005 * \epsilon_t$ $b_t = b_{t-1} + 0.0005 * \epsilon_t$

The model indicates a slightly declining trend, but small beta value (beta=0.0005) suggest the level and trend are relatively stable over time. The model also puts very little weight on recent observations (alpha=0.0005).

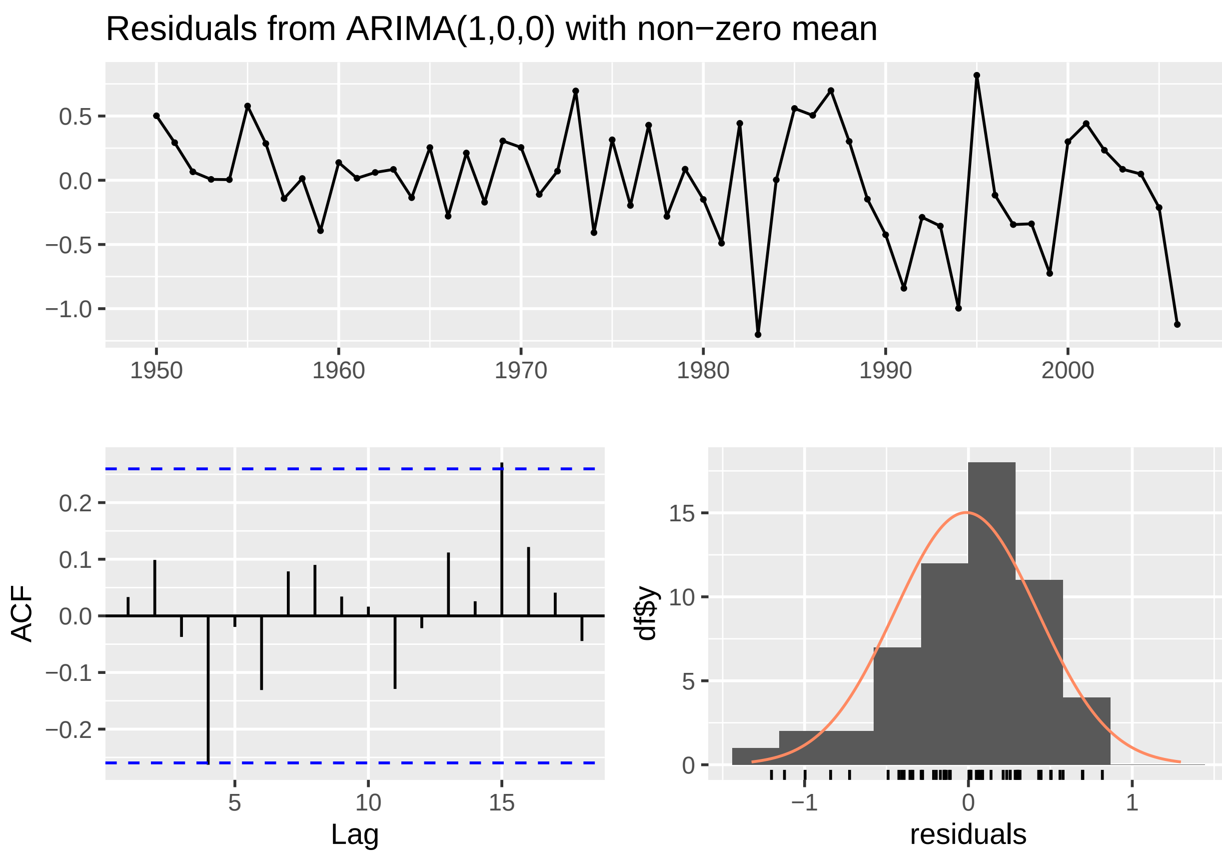
```
##         0,0,0   0,1,0    1,0,0    1,1,0    1,1,1     Holt
## RMSE 3.053917 2.96879 3.048373 3.014439 3.005601 2.881865
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ETS(A,A,N)
## Q* = 37.425, df = 10, p-value = 4.776e-05
##
## Model df: 0.   Total lags used: 10
```

Oregon The best fit model for Oregon was a first-order ARIMA model with no differencing and no moving average. The time series is stationary and auto-correlated around a mean of 7.5190, with observations being moderately dependent on recent values (ar1<1). $y_t = 0.7303 * y_{t-1} + 2.026 + \epsilon_t$ ϵ_t $N(0, \sigma^2 = 0.1876)$

```
##          0,0,0     0,1,0     1,0,0     1,1,0     1,1,1     Holt
## RMSE 0.8713649 0.6326566 0.5259991 0.5739301 0.5861011 0.737802
```

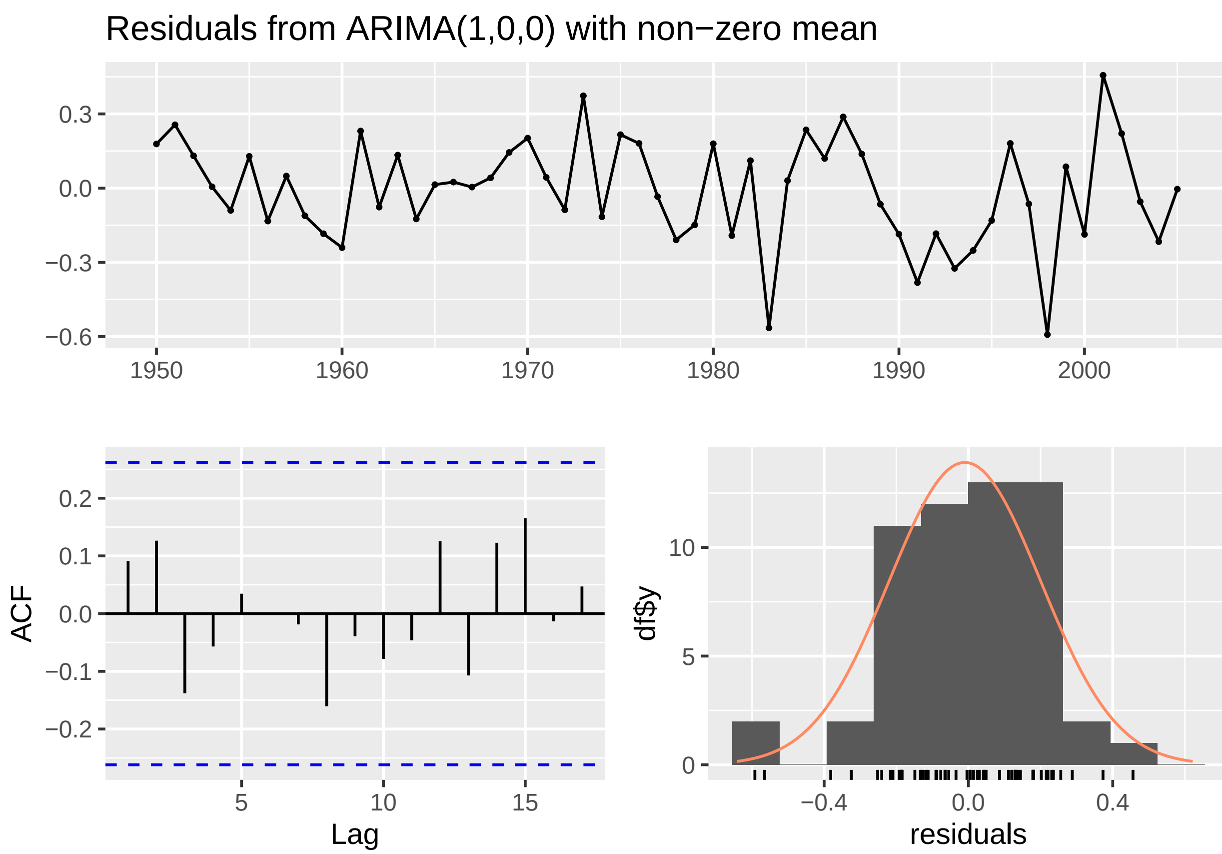



```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(1,0,0) with non-zero mean
## Q* = 7.3609, df = 9, p-value = 0.5996
##
## Model df: 1.   Total lags used: 10
```

Washington Like Oregon, the best fit model for Washington was a first-order ARIMA model with no differencing and not moving average. The values differ slightly, with Washington having a slightly higher average than Oregon (intercept=7.9066), and autoregression component (ar1) of 0.8878, meaning the series shows strong positive correlation.

$y_t = 0.8878 * y_{t-1} + 0.8871 + \epsilon_t \ \epsilon_t \ N(0, \sigma^2 = 0.0436)$

```
##          0,0,0     0,1,0     1,0,0     1,1,0     1,1,1      Holt
## RMSE 0.3707379 0.3193469 0.2504157 0.3188799 0.3192479 0.3744703
```

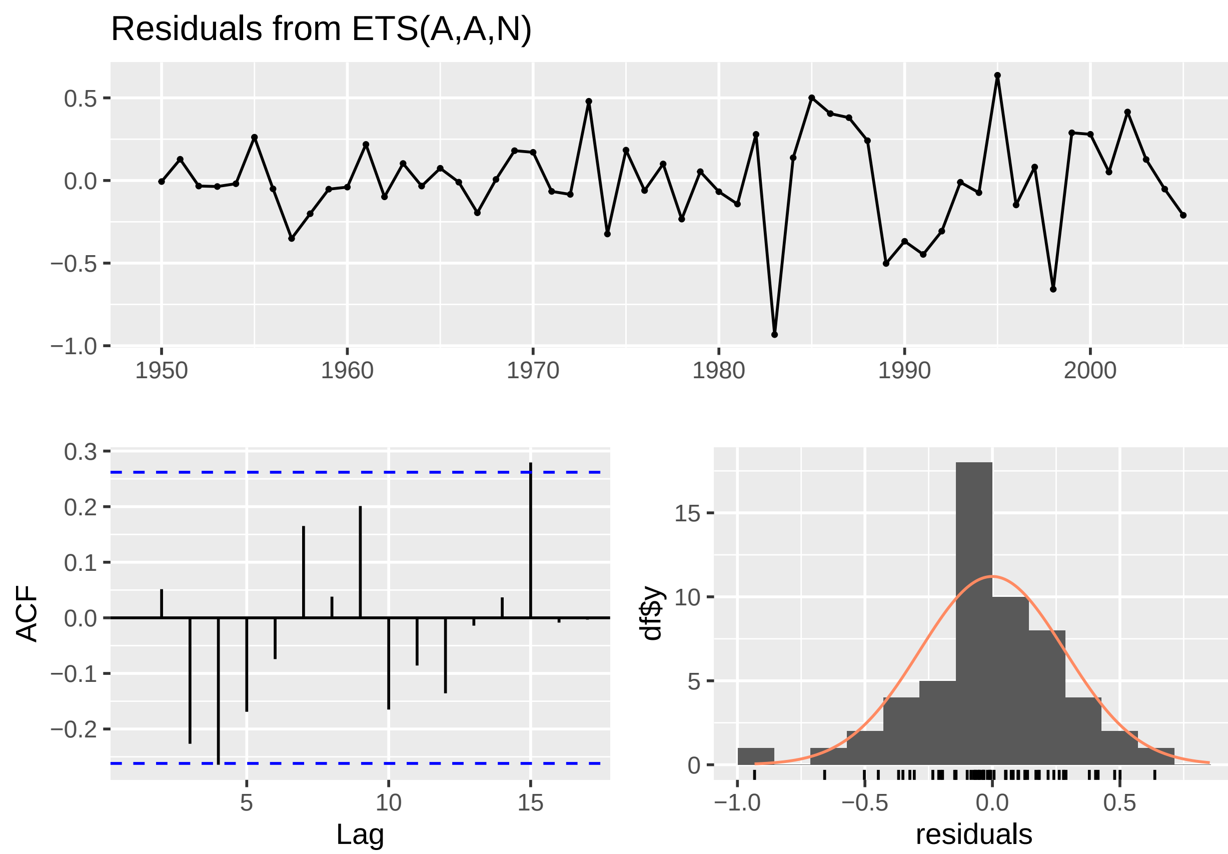



```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(1,0,0) with non-zero mean
## Q* = 5.206, df = 9, p-value = 0.816
##
## Model df: 1.   Total lags used: 10
```

PNW Region The model that best fit the Pacific Northwest region as a whole was a Holt model with the following equation: $y_t = l_{t-1} + b_{t-1} + \epsilon_t$
$l_t = l_{t-1} + b_{t-1} + 0.9619 * \epsilon_t$
$b_t = b_{t-1} + 0.0001 * \epsilon_t$

The small beta value suggests that the trend is stable over time, and the alpha value being close to 1 indicates that the model assigns greater weight to recent observations. The model has identified a slightly negative trend in chinook landings over time (b=-0.0108).

```
##          0,0,0     0,1,0     1,0,0     1,1,0     1,1,1      Holt
## RMSE 0.8135519 0.6220047 0.7505696 0.6286976 0.6289971 0.5934286
```

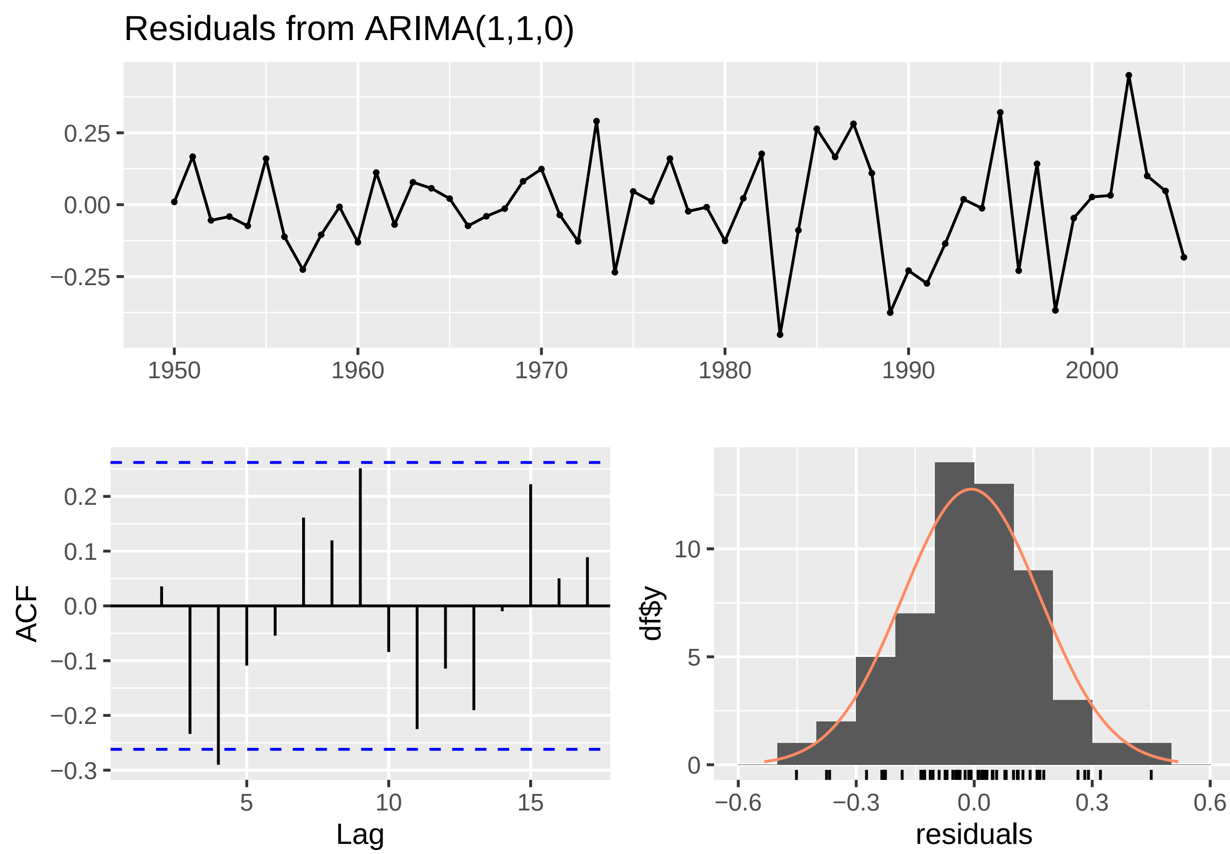



```
##
##  Ljung-Box test
##
## data:  Residuals from ETS(A,A,N)
## Q* = 16.47, df = 10, p-value = 0.08694
##
## Model df: 0.   Total lags used: 10
```

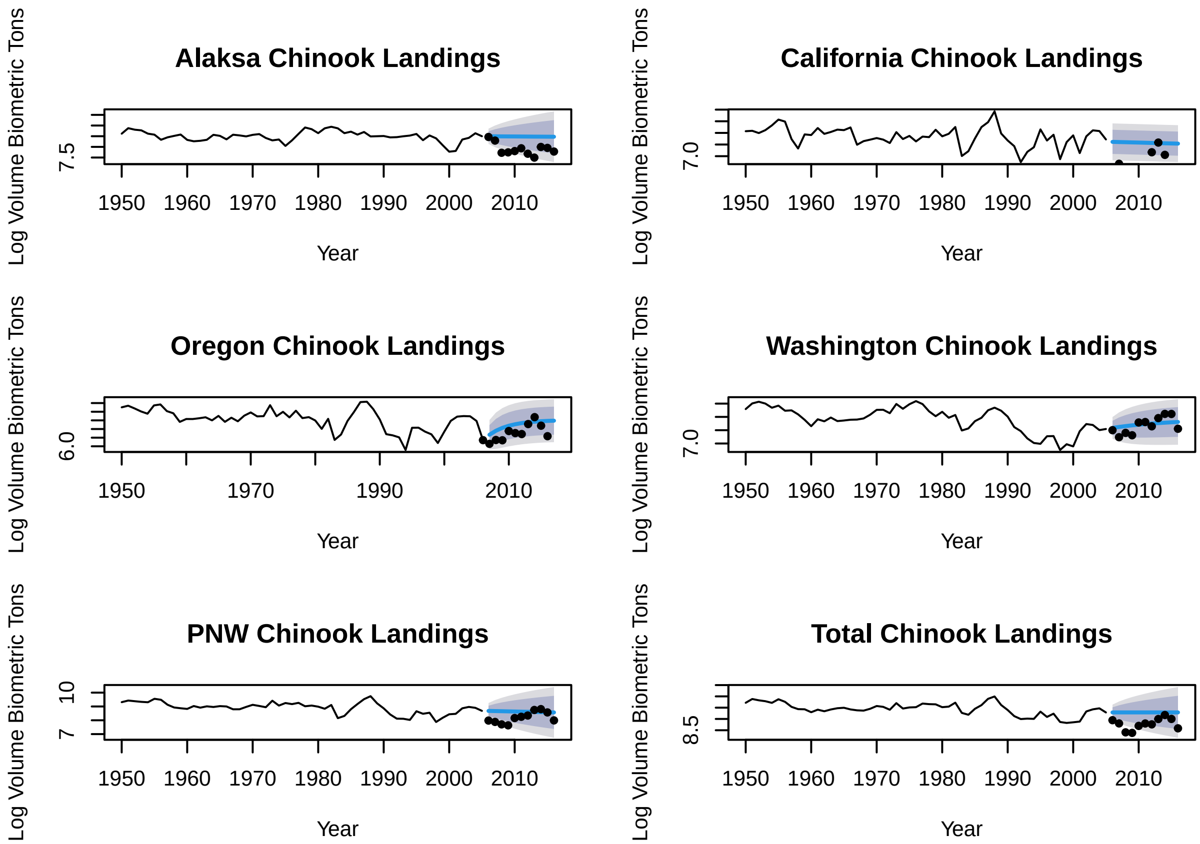
Total chinook landings The best fit model for total chinook landings was a first-order ARIMA with first difference since the data was non-stationary, and no seasonality: $y_t = 0.8878 * y_{t-1} + 0.8871 + \epsilon_t$ $\epsilon_t \; N(0, \sigma^2 = 0.0436)$

The alpha value suggests weak, negative autocorrelation, but it is not statistically significant.

```
##         0,0,0   0,1,0     1,0,0     1,1,0     1,1,1      Holt
## RMSE 0.704873 0.56315 0.6618483 0.5659348 0.5694472 0.5312216
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(1,1,0)
## Q* = 17.185, df = 9, p-value = 0.04589
##
## Model df: 1.   Total lags used: 10
```



In summary, none of these models contained a seasonality component. The model structure differed slightly between regions, but none of the models found significant changes in trend over time. Many of them behaved functionally like a random walk.

Forecasting Accuracy Comparison

When we compared forecasting accuracy between models fit to one region against test data from another region, we found some interesting results. In some cases, another state's forecasted data was a better fit than the best fit model for that state. This was the case for Alaska, whose test data was more accurately forecasted by the ARIMA model fit to the Washington data. Only the Washington and Oregon model forecasts were the best fit for their state's test data. This is the opposite of what I would have expected, given our original hypothesis that Washington and Oregon's time series look very similar. I would have expected forecasts from these two states to not be accurate when tested against other states' data, since California and Alaska's data looked different, but the Washington and Oregon models performed fairly well across the board. The large regional models (PNW and total) did not perform well, which does support the idea that forecasts for salmon fisheries should be made on smaller regional scales that are ecologically relevant to salmon populations.

```
##                 AK Model  CA Model  OR Model  WA Model PNW Model Total Model
## AK Forecast    0.6447745 0.4089038 0.7335605 0.3607922 0.7691459   1.4183031
## CA Forecast    3.5268146 2.8818654 2.7096443 2.9523386 3.6400794   4.1818441
## OR Forecast    1.7887483 0.9452960 0.5259991 1.0418512 1.9274951   2.5737358
## WA Forecast    0.8638981 0.3241475 0.5095812 0.2504157 1.0009427   1.6417700
## PNW Forecast   0.4860730 0.7278896 1.0296953 0.5795634 0.5934286   1.1656268
## Total Forecast 0.3771229 1.2245373 1.5626548 1.0840916 0.2934693   0.5659348
```

Question 2

The Forecast package's auto.arima function was used to determine which ARIMA models best fit the training data related to landings, prices, and value of fish landed.

```
## Series: CAlands.train
## ARIMA(5,1,0) with drift
##
## Coefficients:
##           ar1     ar2     ar3      ar4      ar5    drift
##       -0.2649  0.3924  0.3555  -0.2716  -0.8196  -0.0031
## s.e.   0.0350  0.0361  0.0371   0.0348   0.0363   0.0358
##
## sigma^2 = 1.238:  log likelihood = -339.5
## AIC=693   AICc=693.31   BIC=720.32
```

```
## Series: CAprice.train
## ARIMA(0,1,1) with drift
##
## Coefficients:
##           ma1    drift
##       -0.5681  28.4341
## s.e.   0.1053  46.4810
##
## sigma^2 = 4256172:  log likelihood = -1616.07
## AIC=3238.13   AICc=3238.2   BIC=3249.84
```

```
## Series: CAvalue.train
## ARIMA(5,0,0) with non-zero mean
##
## Coefficients:
##          ar1     ar2      ar3      ar4      ar5      mean
##       0.2534  0.3793  -0.0226  -0.2175  -0.3965  816239.5
## s.e.  0.0794  0.0777   0.0913   0.0953   0.0852  115365.9
##
## sigma^2 = 1.01e+12:  log likelihood = -2762.04
## AIC=5538.09   AICc=5538.4   BIC=5565.42
```

```
## Series: ORlands.train
## ARIMA(4,0,0)(0,1,0)[12]
##
## Coefficients:
##          ar1     ar2     ar3     ar4
##       0.3130  0.1072  0.0666  0.0113
## s.e.  0.0716  0.0878  0.0761  0.0636
##
## sigma^2 = 0.9453:  log likelihood = -405.48
## AIC=820.96   AICc=821.13   BIC=840.36
```

```
## Series: ORprice.train
## ARIMA(3,0,0)(0,1,0)[12]
##
## Coefficients:
```

```
##          ar1      ar2     ar3
##       0.5521  -0.2566  0.1190
## s.e.  0.0822   0.0906  0.0687
##
## sigma^2 = 9604822:  log likelihood = -2732.55
## AIC=5473.11   AICc=5473.22   BIC=5488.63
```

```
## Series: ORvalue.train
## ARIMA(0,1,0)(1,0,0)[12]
##
## Coefficients:
##         sar1
##       0.4295
## s.e.  0.0509
##
## sigma^2 = 4.122e+11:  log likelihood = -4501.94
## AIC=9007.88   AICc=9007.91   BIC=9015.7
```

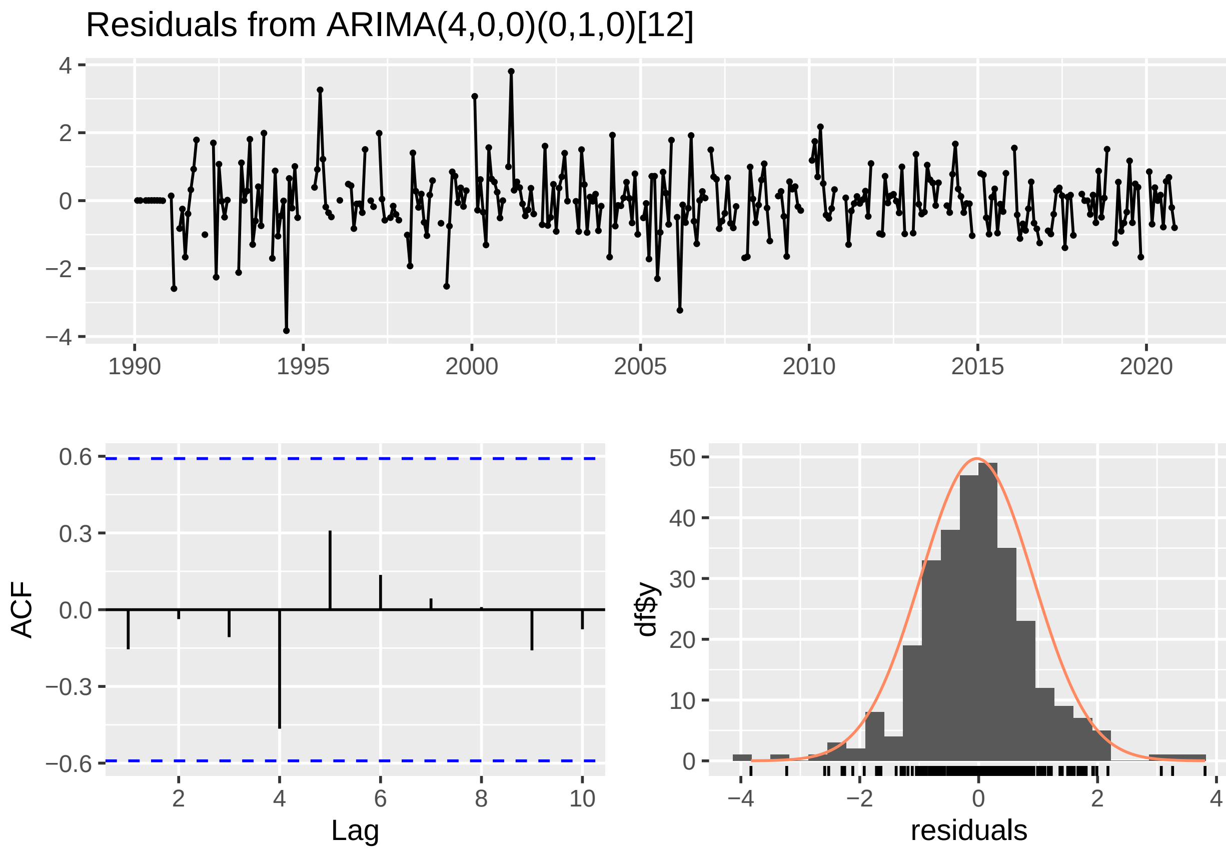
```
## Series: WAlands.train
## ARIMA(0,0,4)(0,1,2)[12] with drift
##
## Coefficients:
##          ma1     ma2     ma3     ma4     sma1     sma2    drift
##       0.3208  0.2465  0.1630  0.1747  -0.5187  -0.2200  -0.0043
## s.e.  0.0591  0.0601  0.0564  0.0580   0.0628   0.0671   0.0024
##
## sigma^2 = 0.8263:  log likelihood = -455.77
## AIC=927.54   AICc=927.95   BIC=958.61
```

```
## Series: WAprice.train
## ARIMA(0,0,0)(2,1,2)[12] with drift
##
## Coefficients:
##          sar1   sar2     sma1     sma2    drift
##       -0.2499  0.275  -0.1082  -0.3960  18.5989
## s.e.   0.3593  0.143   0.3559   0.1972   4.9433
##
## sigma^2 = 4071435:  log likelihood = -3088.97
## AIC=6189.93   AICc=6190.17   BIC=6213.23
```

```
## Series: WAvalue.train
## ARIMA(0,0,3)(2,1,0)[12] with drift
##
## Coefficients:
##          ma1     ma2      ma3     sar1     sar2      drift
##       0.1216  0.1652  -0.0179  -0.2615  -0.0113  -517.2173
## s.e.  0.0583  0.0490   0.0530   0.0556   0.0606  2020.5305
##
## sigma^2 = 2.116e+11:  log likelihood = -4959.7
## AIC=9933.4   AICc=9933.72   BIC=9960.59
```

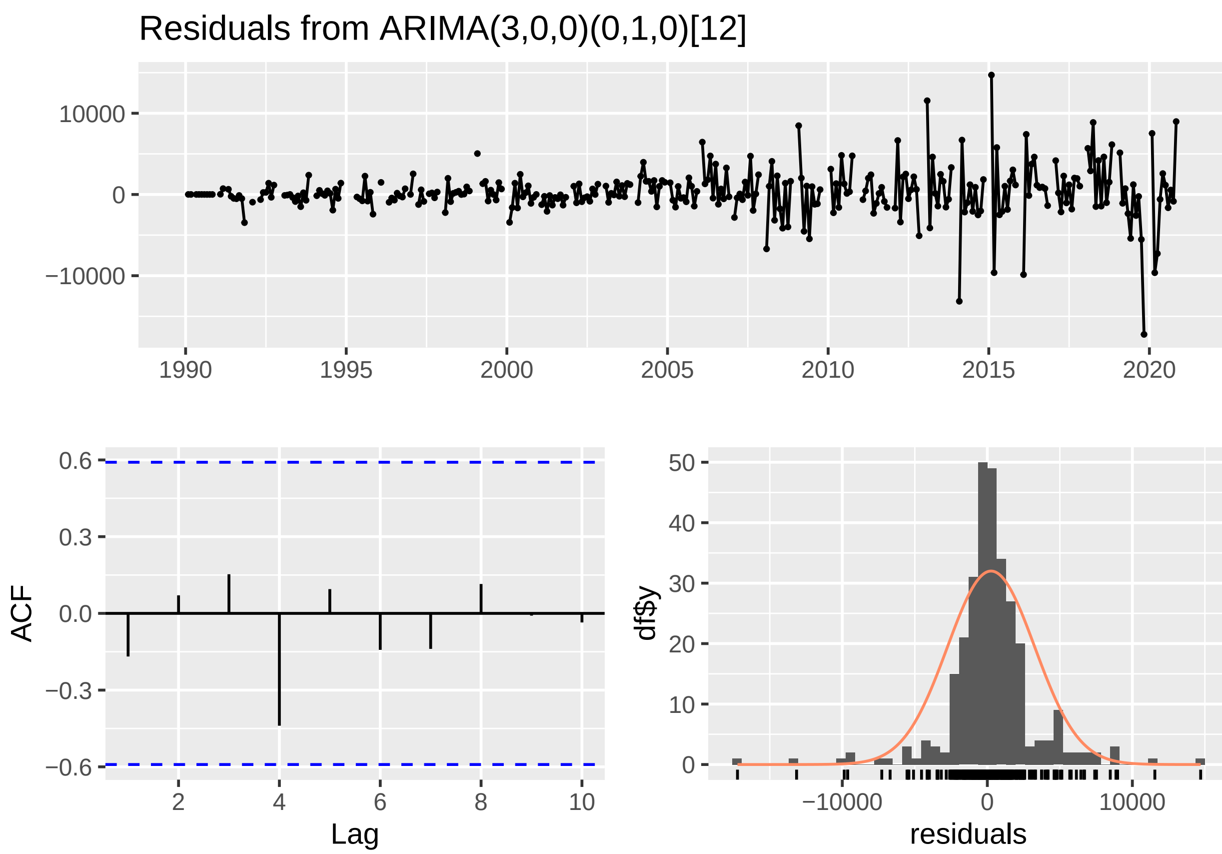
The models describing Chinook market data in California fail to find any seasonal trends, most likely due to data from California being much more scarce than data from Oregon and Washington. Moving forward California will be omitted from the process of forecasting landings, prices, and value.

```r
checkresiduals(ORlands.fit)
```



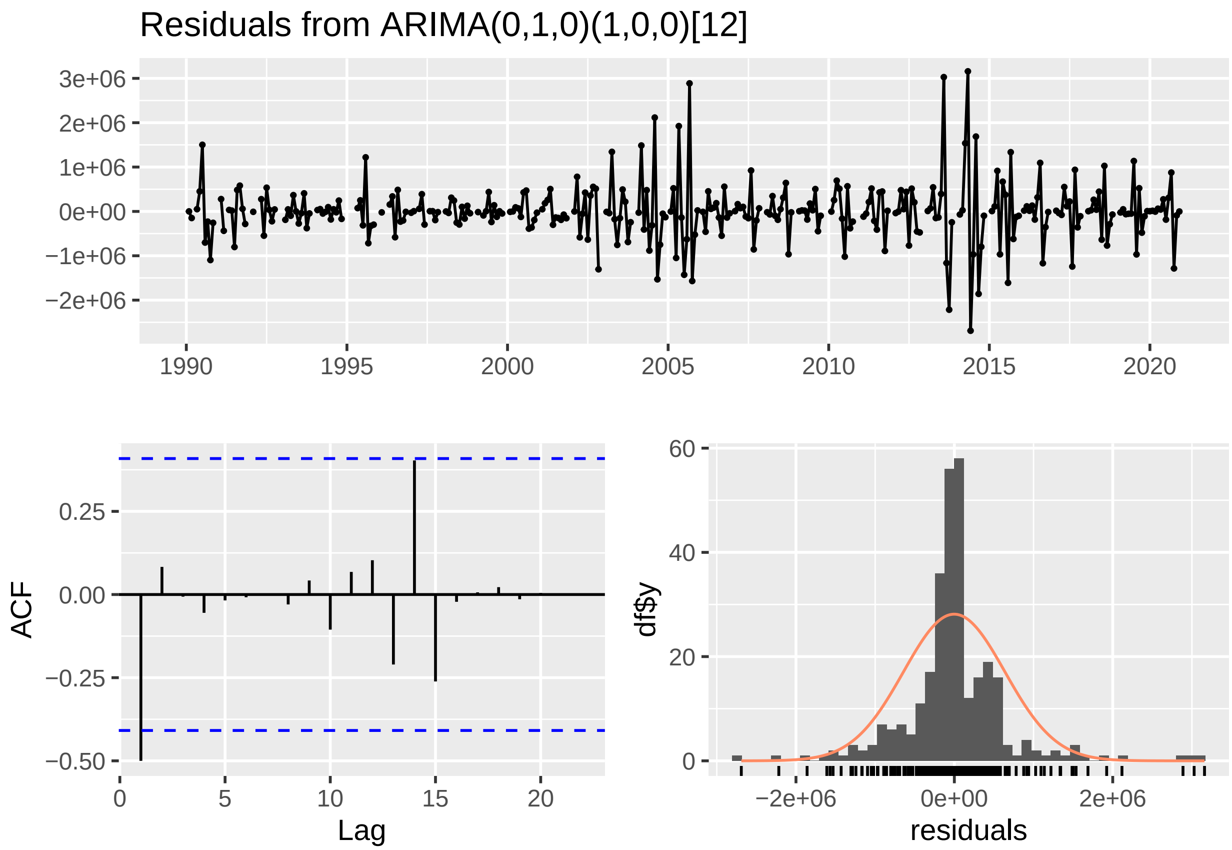

```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(4,0,0)(0,1,0)[12]
## Q* = 70.352, df = 20, p-value = 1.596e-07
##
## Model df: 4.   Total lags used: 24
```

```r
checkresiduals(ORprice.fit)
```



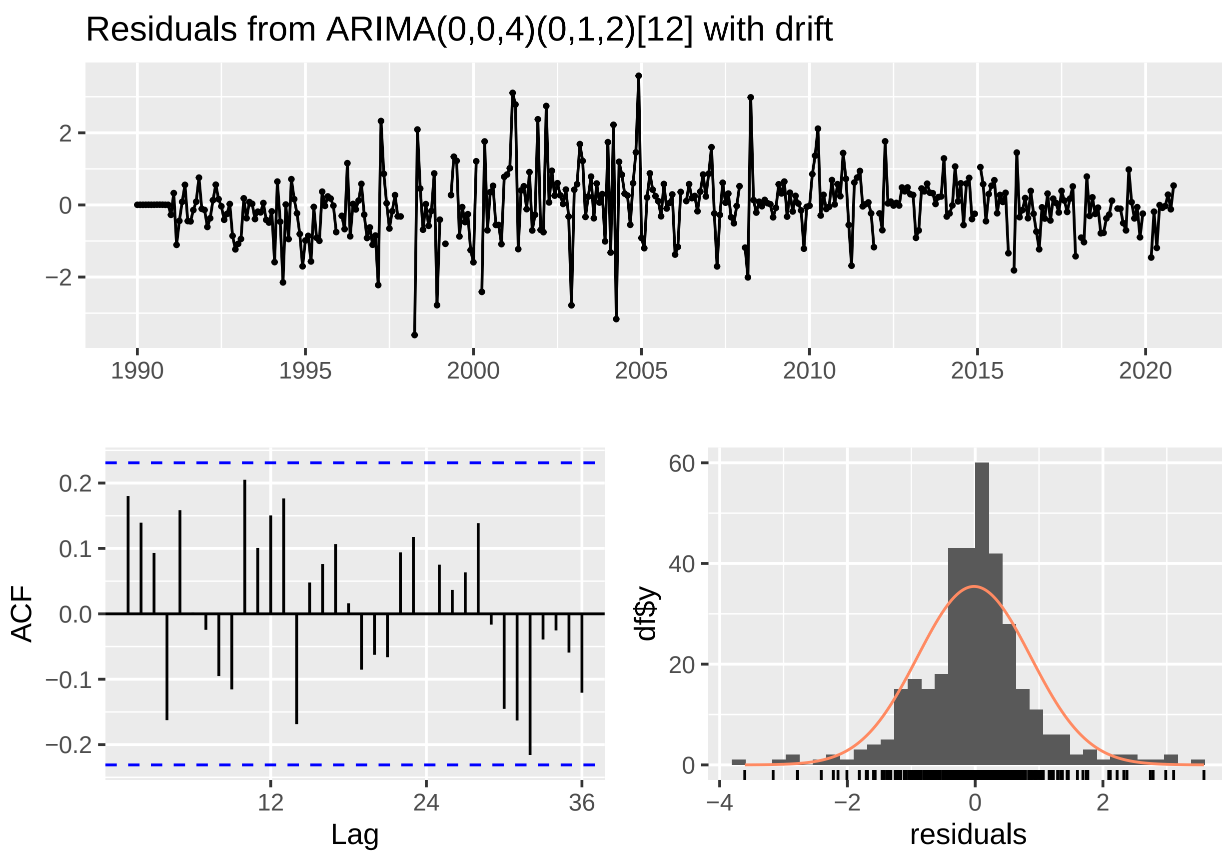

```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(3,0,0)(0,1,0)[12]
## Q* = 113.96, df = 21, p-value = 8.993e-15
##
## Model df: 3.   Total lags used: 24
```

```r
checkresiduals(ORvalue.fit)
```



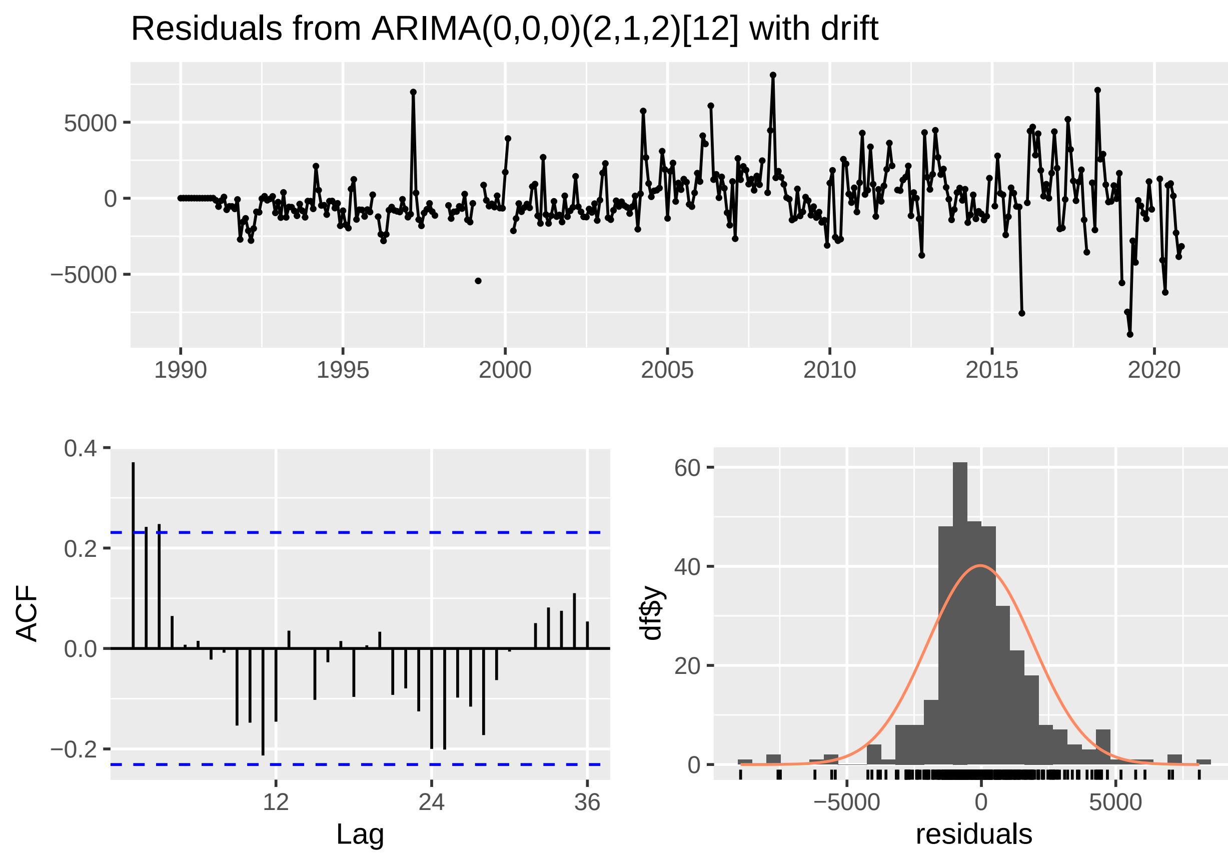

```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,1,0)(1,0,0)[12]
## Q* = 84.653, df = 23, p-value = 5.512e-09
##
## Model df: 1.   Total lags used: 24
```

```r
checkresiduals(WAlands.fit)
```



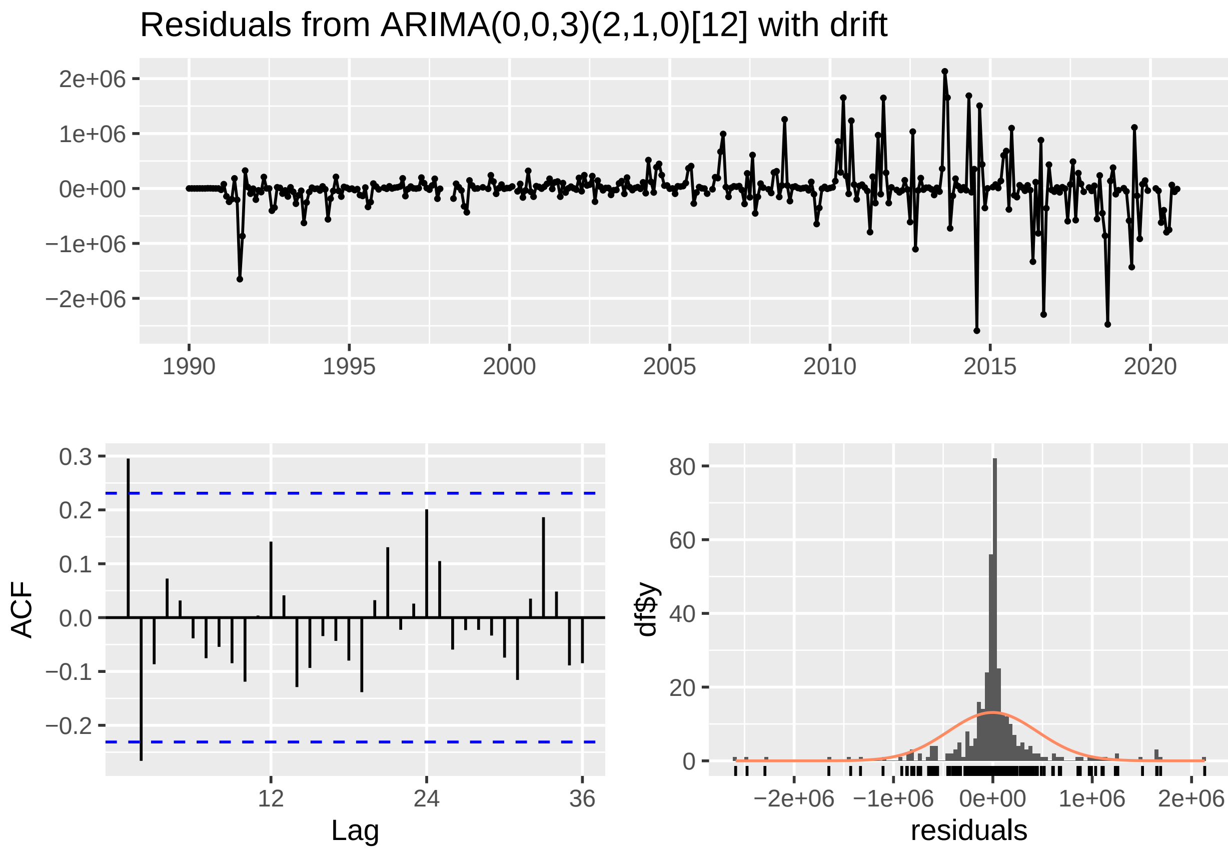

```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,0,4)(0,1,2)[12] with drift
## Q* = 53.179, df = 18, p-value = 2.461e-05
##
## Model df: 6.   Total lags used: 24
```

```r
checkresiduals(WAprice.fit)
```



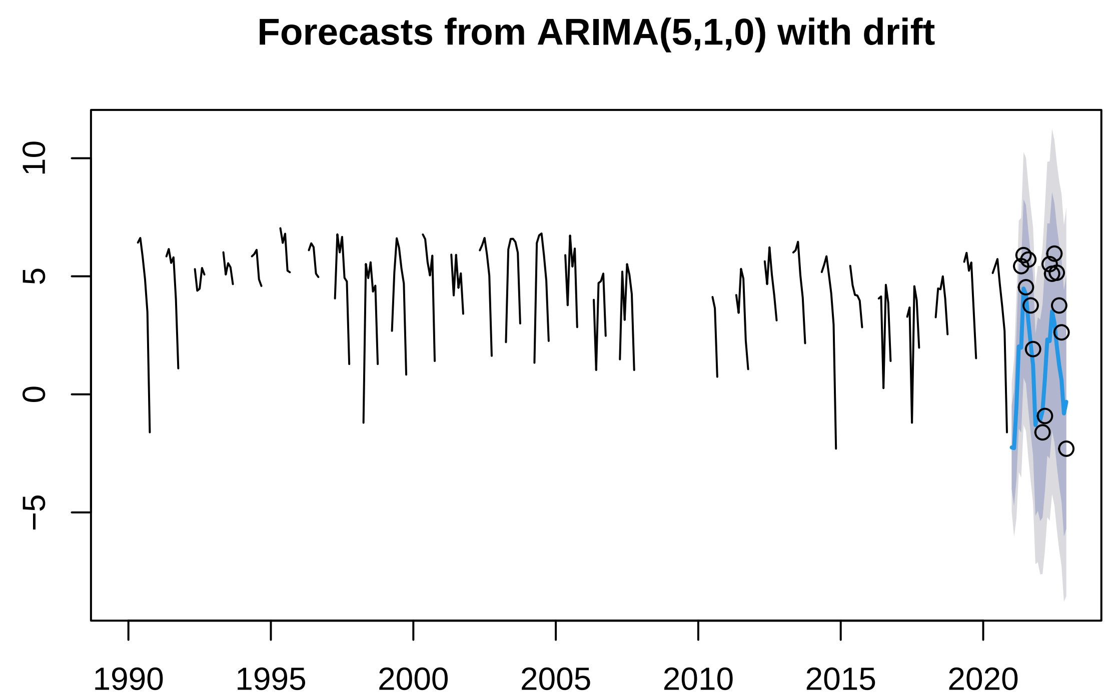

```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,0,0)(2,1,2)[12] with drift
## Q* = 201.45, df = 20, p-value < 2.2e-16
##
## Model df: 4.   Total lags used: 24
```

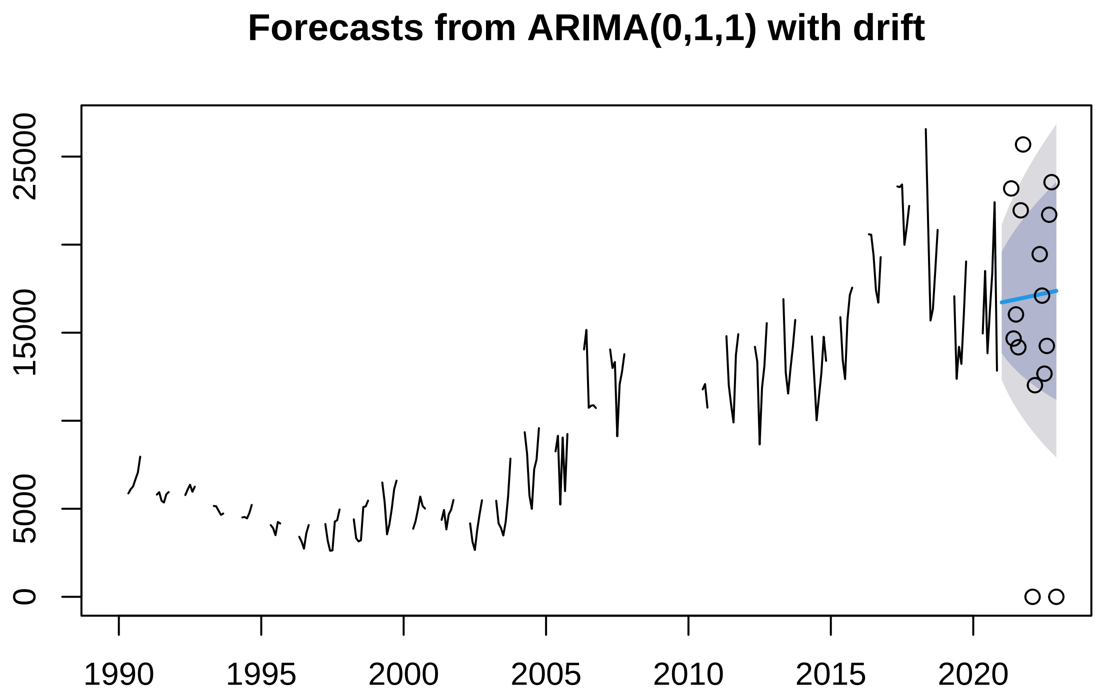
```r
checkresiduals(WAvalue.fit)
```



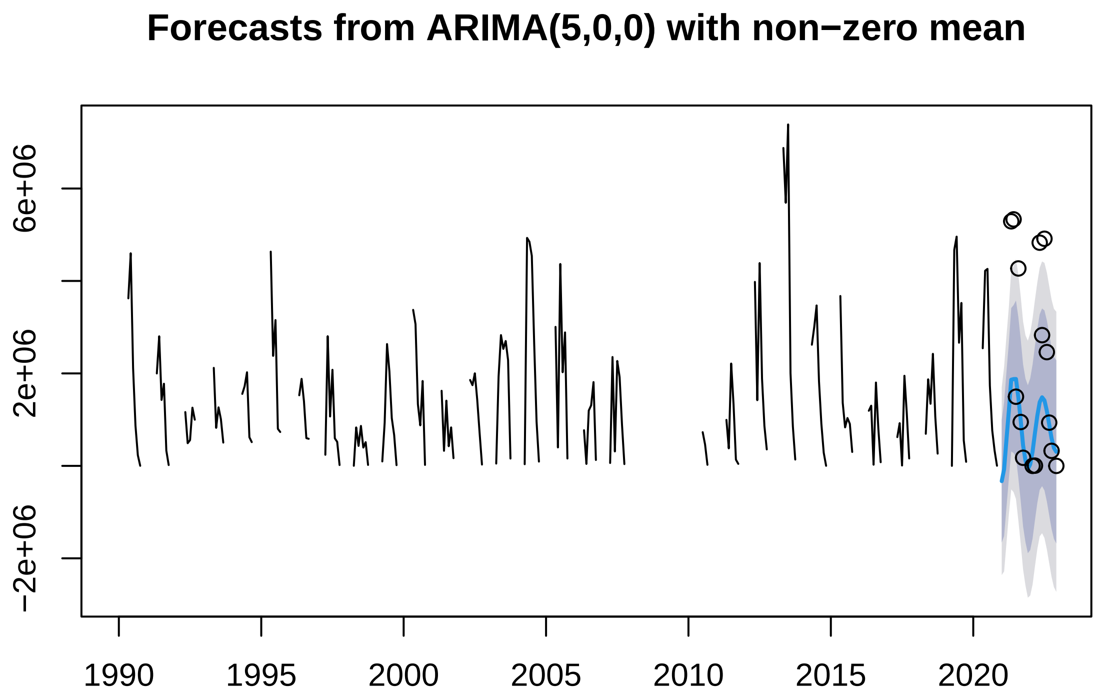

```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,0,3)(2,1,0)[12] with drift
## Q* = 62.201, df = 19, p-value = 1.727e-06
##
## Model df: 5.   Total lags used: 24
```

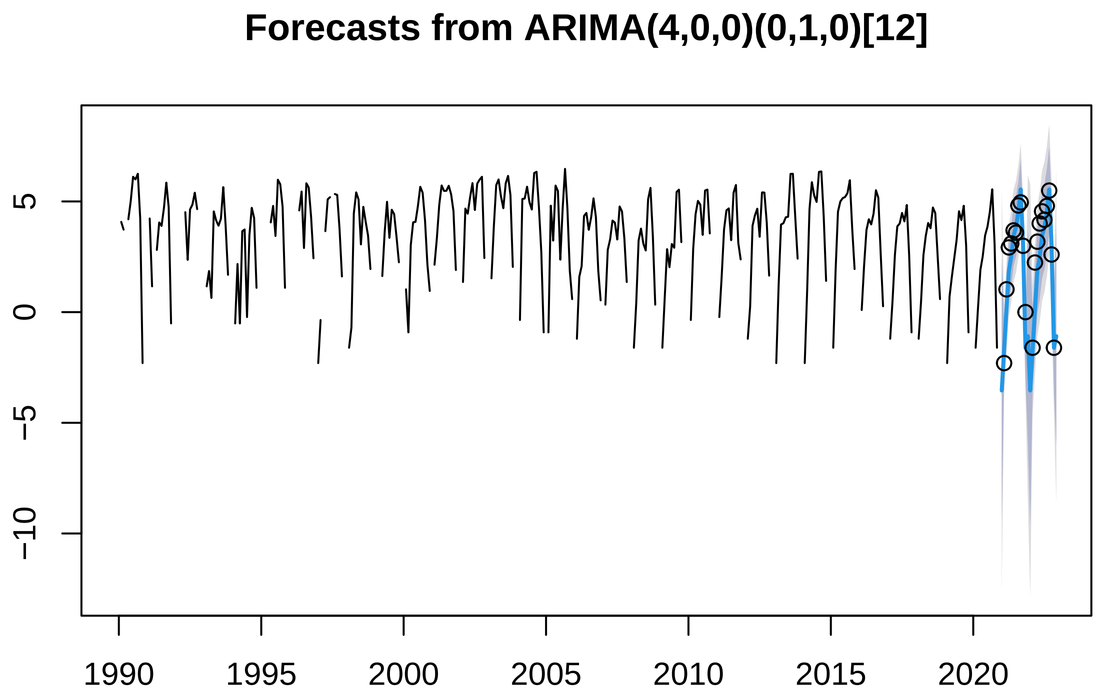
In general, models fitting landed weight, prices, and value show well behaved residuals distributed symmetrically around a mean of zero with little auto-correlation. The value models for both Oregon and Washington show the most autocorrelation and also a very tight distribution of error terms.

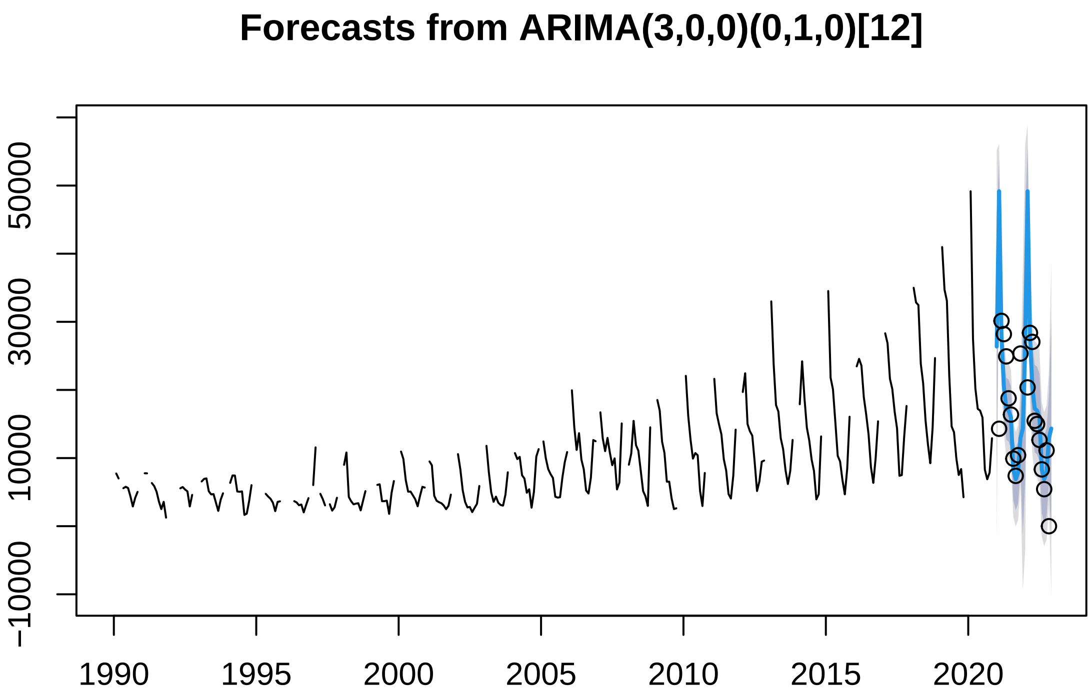

Forecasts from ARIMA(5,1,0) with drift
10
5
0
−5
1990
1995
2000
2005
2010
2015
2020

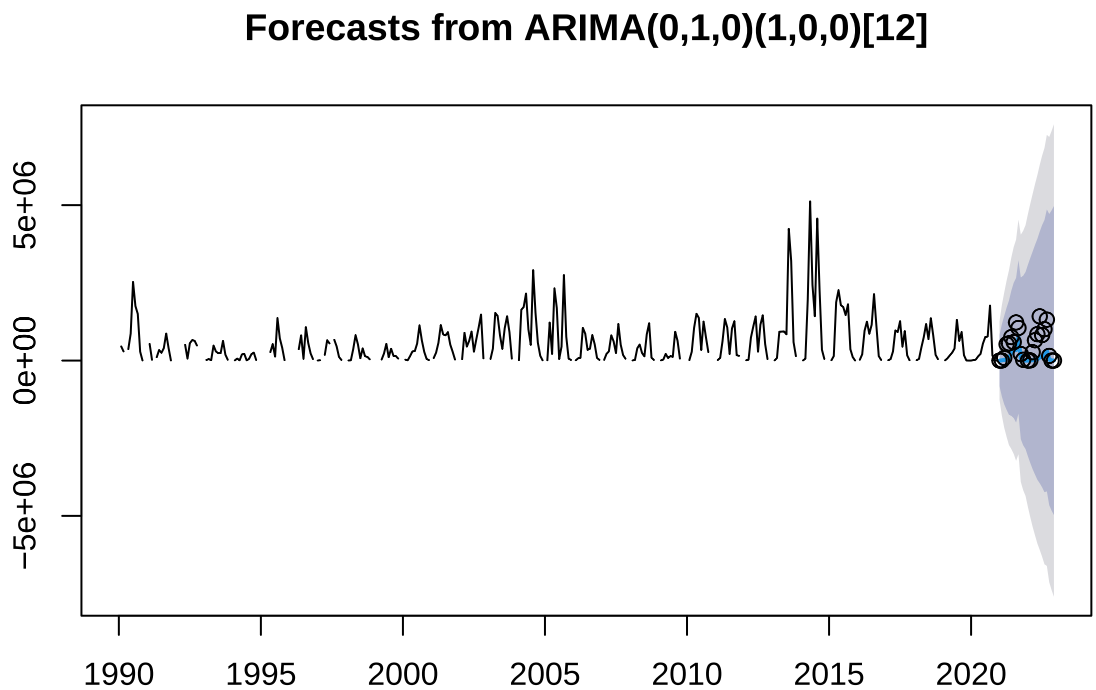


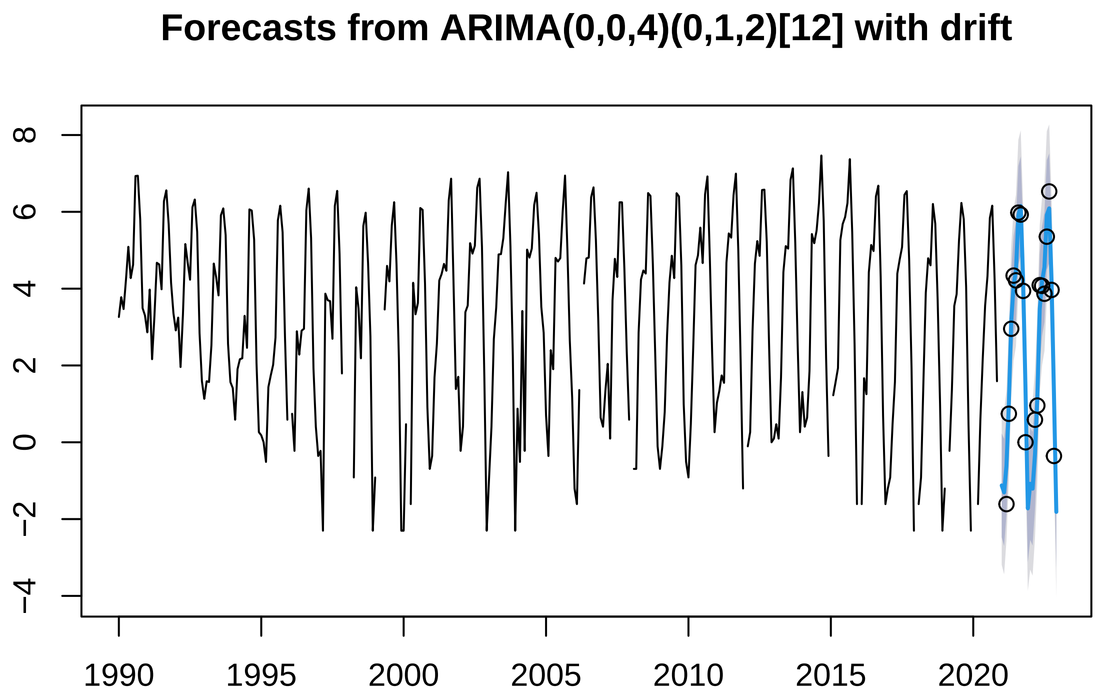
Forecasts from ARIMA(0,1,1) with drift
25000
15000
5000
0
1990
1995
2000
2005
2010
2015
2020


Forecasts from ARIMA(5,0,0) with non−zero mean
6e+06
2e+06
−2e+06
1990
1995
2000
2005
2010
2015
2020

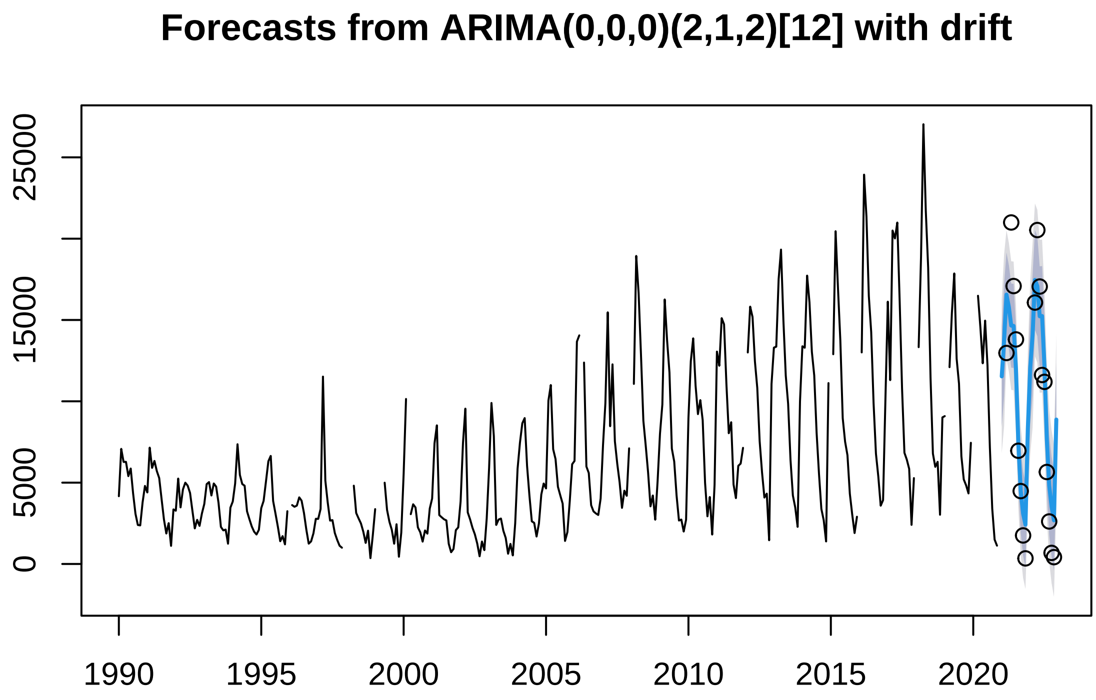

Forecasts from ARIMA(4,0,0)(0,1,0)[12]
5
0
−5
−10
1990
1995
2000
2005
2010
2015
2020


Forecasts from ARIMA(3,0,0)(0,1,0)[12]
−10000
10000
30000
50000
1990
1995
2000
2005
2010
2015
2020

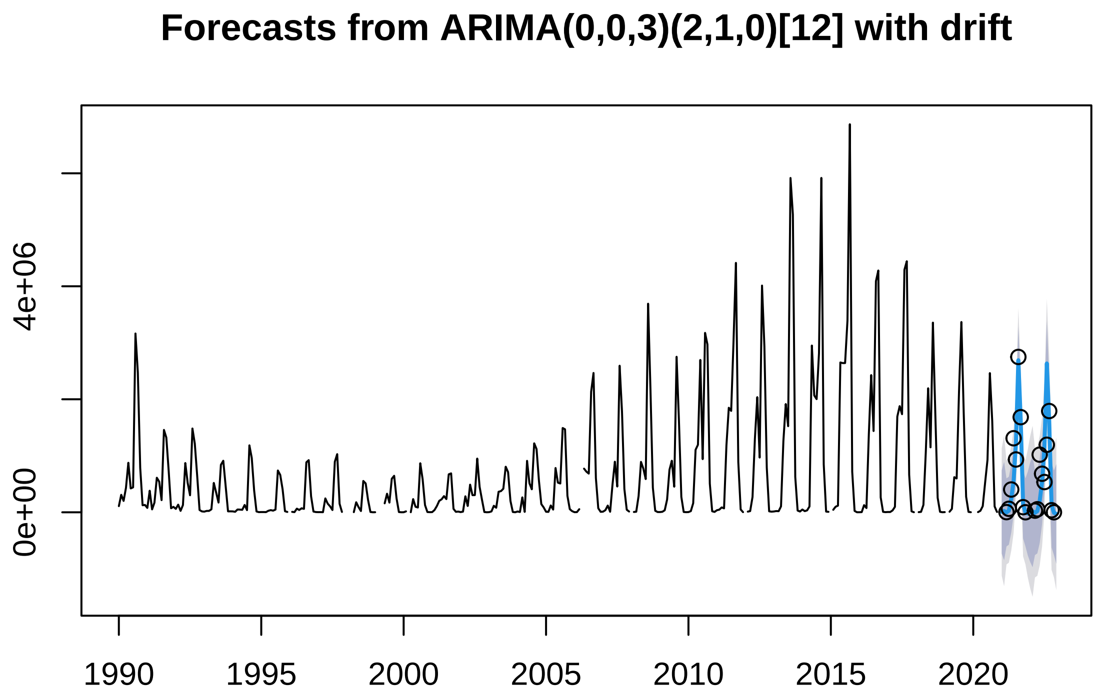

Forecasts from ARIMA(0,1,0)(1,0,0)[12]
5e+06
0e+00
−5e+06
1990
1995
2000
2005
2010
2015
2020



Forecasts from ARIMA(0,0,4)(0,1,2)[12] with drift
8
6
4
2
0
−2
−4
1990
1995
2000
2005
2010
2015
2020



Forecasts from ARIMA(0,0,0)(2,1,2)[12] with drift
0
5000
15000
25000
1990
1995
2000
2005
2010
2015
2020



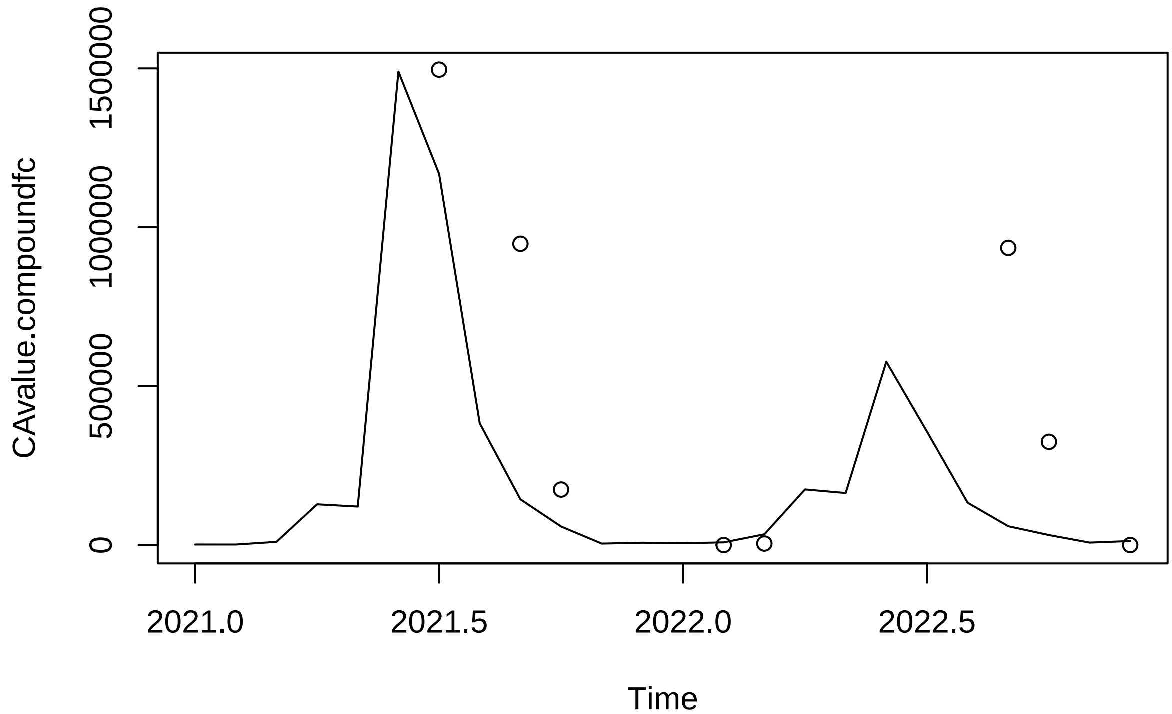

```
##              Jan         Feb         Mar         Apr         May         Jun
## 2021    1766.045    1717.487   10087.195  128282.436  121191.852 1489916.937
## 2022    5698.177    8479.606   33832.115  175085.026  163739.387  576865.677
##              Jul         Aug         Sep         Oct         Nov         Dec
## 2021 1168323.143  383081.701  143907.902   58408.212    4631.498    7405.420
## 2022  357166.145  133311.382   59256.253   31502.462    7725.489   12665.582
```

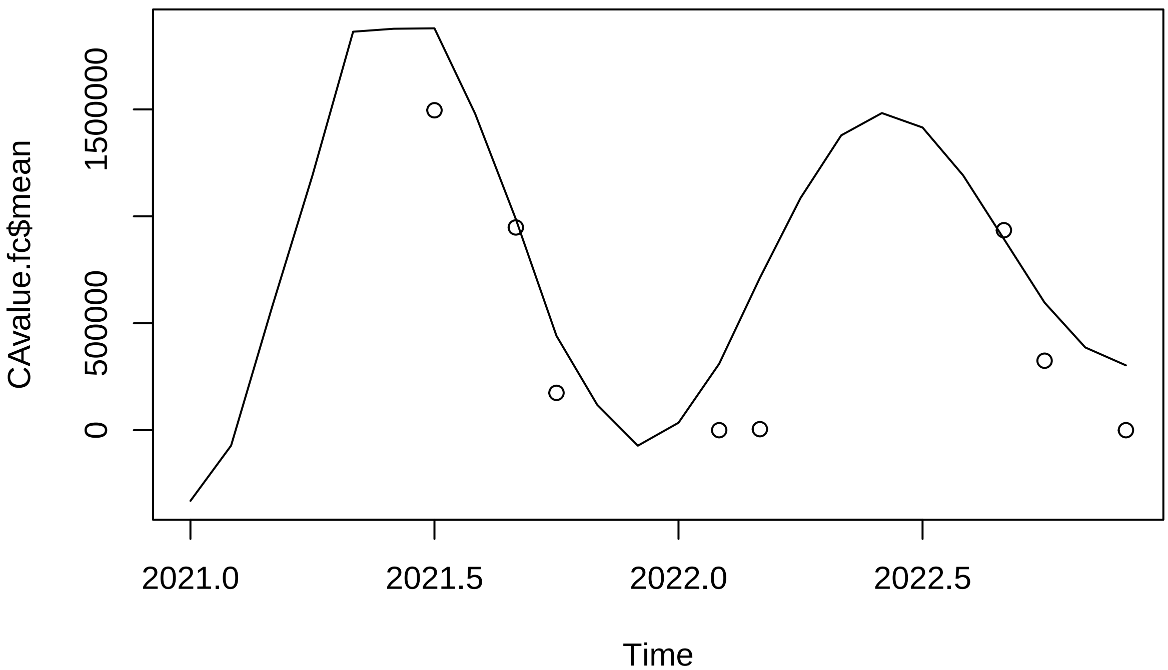
```
##             Jan        Feb        Mar        Apr        May        Jun
## 2021 -330499.91  -71961.78  571354.34 1191308.00 1863549.81 1877302.35
## 2022   34783.29  310730.66  712180.26 1085515.21 1378759.46 1483071.86
##             Jul        Aug        Sep        Oct        Nov        Dec
## 2021 1879316.37 1479914.49  987140.02  441185.70  119077.33  -72410.84
## 2022 1415559.15 1191006.77  894328.73  596543.29  386956.78  303216.53
```

```
##          Jan     Feb     Mar     Apr     May     Jun     Jul     Aug     Sep
## 2021      NA      NA      NA      NA 5289926 5332039 1496173 4270173  948128
## 2022      NA       0    4807      NA 4827657 2827524 4913030 2460746  935203
##          Oct     Nov     Dec
## 2021  174684      NA      NA
## 2022  324938      NA       0
```

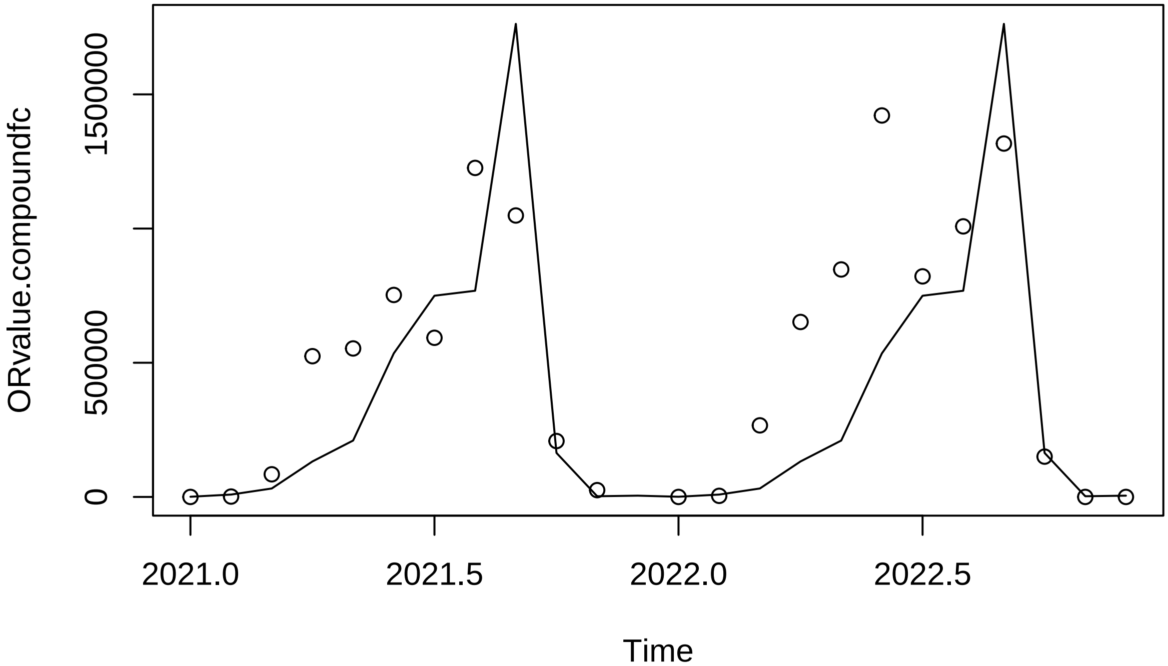
```
##              Jan          Feb          Mar          Apr          May
## 2021    763.7498    8907.2915   31546.9384  132292.4474  210135.3208
## 2022    762.9858    8901.3066   31532.9329  132253.6353  210094.1801
##              Jun          Jul          Aug          Sep          Oct
```

```
## 2021  535083.9774  749792.3051  768314.6258 1762754.5841  164008.1487
## 2022  535014.0503  749727.3127  768270.6448 1762687.3840  164003.9772
##               Nov          Dec
## 2021    2577.3747    4842.7992
## 2022    2577.3312    4842.7449

##               Jan          Feb          Mar          Apr          May
## 2021 5.026625e-21 4.223302e+03 1.416672e+04 5.821267e+04 9.244920e+04
## 2022 2.182787e-11 1.813920e+03 6.084648e+03 2.500251e+04 3.970720e+04
##               Jun          Jul          Aug          Sep          Oct
## 2021 2.339038e+05 3.253132e+05 3.318124e+05 7.605826e+05 7.068071e+04
## 2022 1.004624e+05 1.397230e+05 1.425144e+05 3.266724e+05 3.035757e+04
##               Nov          Dec
## 2021 1.109406e+03 2.182787e-11
## 2022 4.764930e+02 2.808065e-11

##          Jan     Feb     Mar     Apr     May     Jun     Jul     Aug     Sep
## 2021       0    1430   84376  524459  553316  752600  593008 1225958 1048715
## 2022       0    4074  266746  651968  847601 1421344  821889 1008140 1316749
##          Oct     Nov     Dec
## 2021  208332   25338      NA
## 2022  150249       0       0

##              Jan         Feb         Mar         Apr         May         Jun
## 2021    3750.177    3729.012    9022.271   43099.757  337755.878  870863.201
## 2022    4100.162    4195.559   12707.273   53100.736  486915.000 1011016.257
##              Jul         Aug         Sep         Oct         Nov         Dec
## 2021 1049222.686 2715584.632 2110525.288  214660.730   10124.636    1489.240
## 2022 1164235.162 2838714.429 2139841.456  234404.854    9463.864    1449.557

##              Jan         Feb         Mar         Apr         May         Jun
## 2021    5973.080   23025.266   -2957.303   27980.935  239790.337  554453.450
## 2022    1295.948   17192.548   -9149.202   20002.370  203391.372  539716.879
##              Jul         Aug         Sep         Oct         Nov         Dec
## 2021 1224094.610 2691790.859 1633688.285  140138.018   -1997.156  -22186.608
## 2022 1152415.351 2634398.978 1620829.076  124584.322   -7917.084  -29182.518

##          Jan     Feb     Mar     Apr     May     Jun     Jul     Aug     Sep
## 2021      NA      NA    2594   61630  403108 1310533  934616 2749640 1683775
## 2022      NA      NA   28930   53378 1019982  682074  535147 1195148 1792024
##          Oct     Nov     Dec
## 2021   90321     343      NA
## 2022   35896     293      NA
```

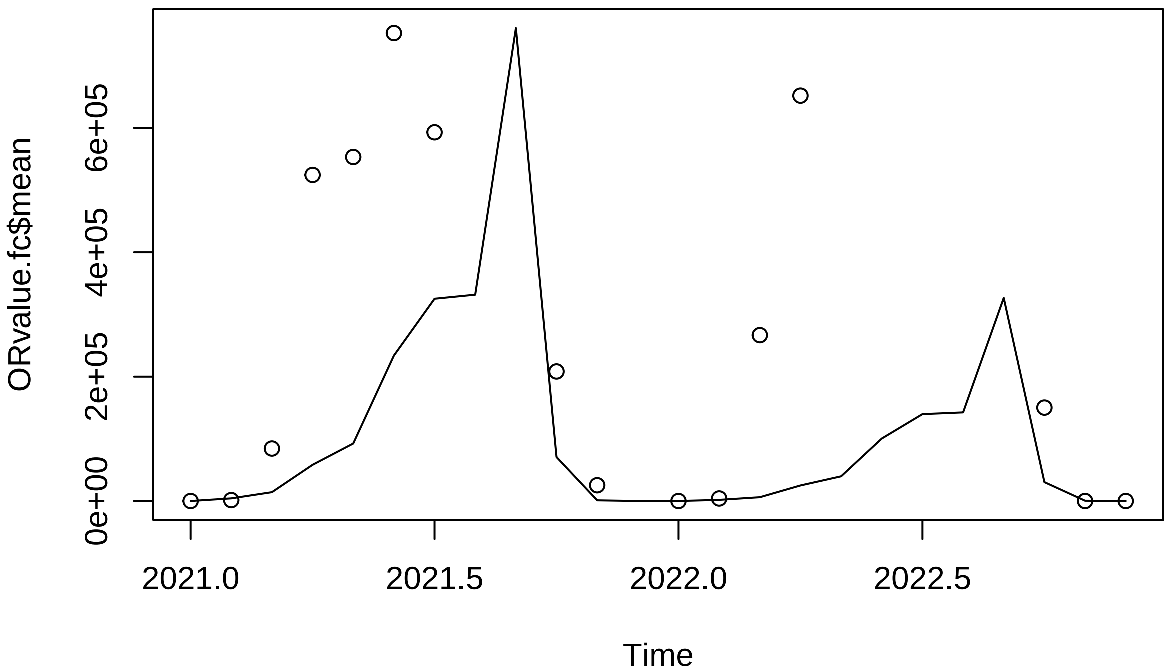

CAvalue.compoundfc
0
500000
1000000
1500000
2021.0
2021.5
2022.0
2022.5
Time



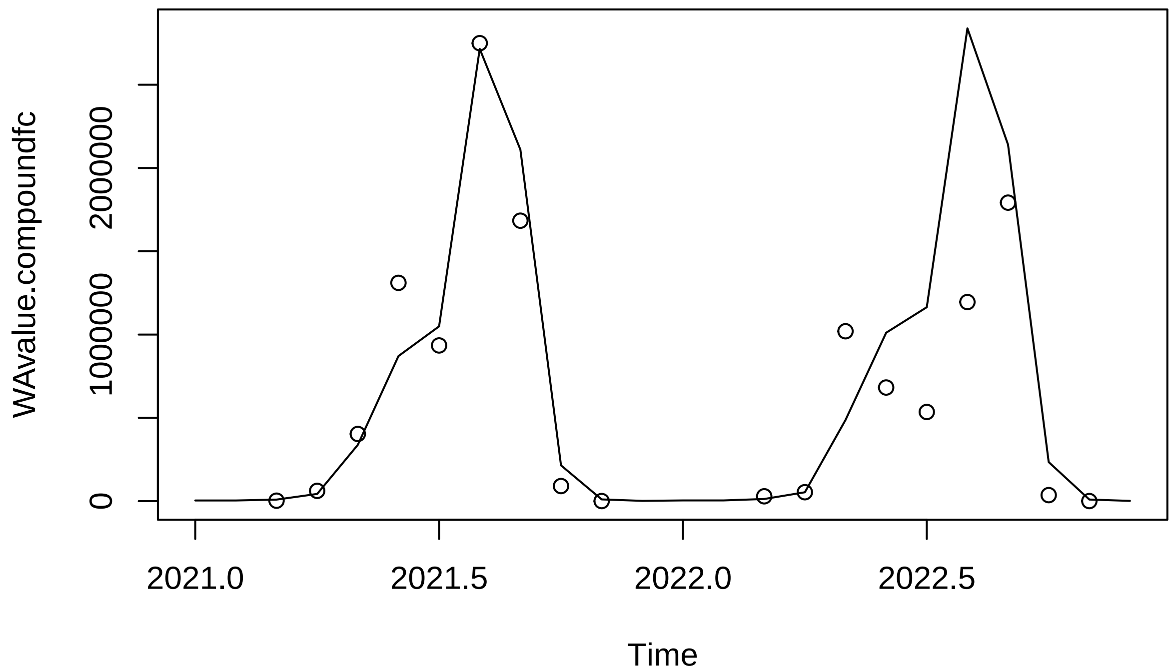
CAvalue.fc$mean
0
500000
1500000
2021.0
2021.5
2022.0
2022.5
Time

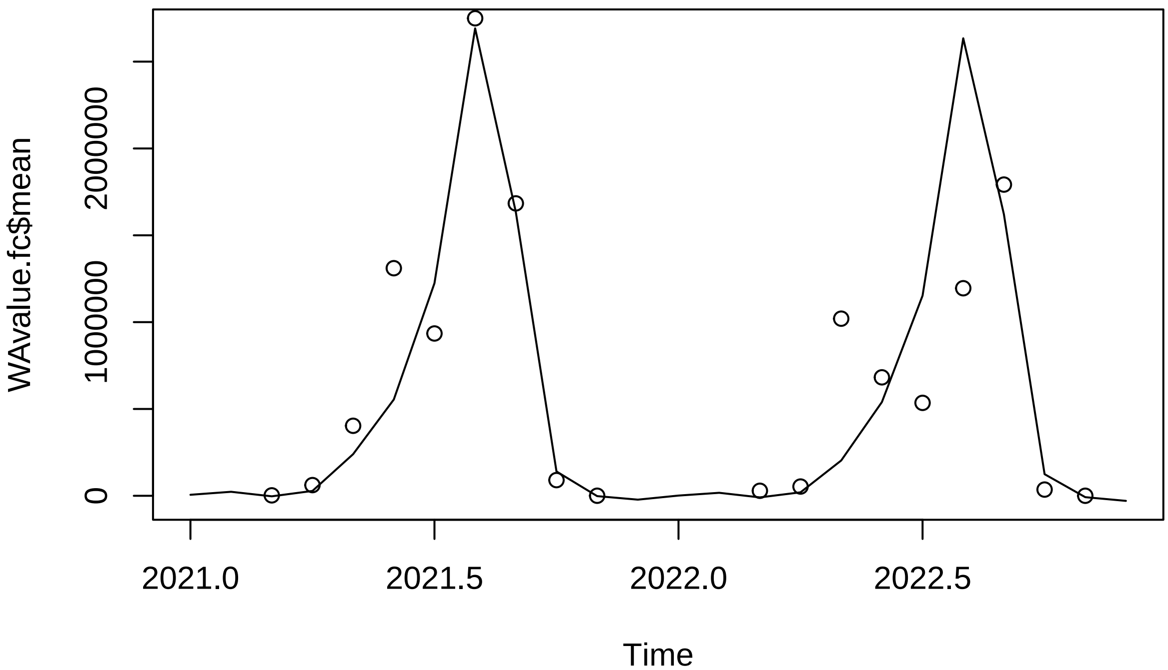


ORvalue.compoundfc
0
500000
1500000
2021.0
2021.5
2022.0
2022.5
Time



ORvalue.fc$mean
0e+00
2e+05
4e+05
6e+05
2021.0
2021.5
2022.0
2022.5
Time


WAvalue.compoundfc
0
1000000
2000000
2021.0
2021.5
2022.0
2022.5
Time



Discussion

In summary, none of these models contained a seasonality component. The model structure differed slightly between regions, but none of the models found significant changes in trend over time. Many of them behaved functionally like a random walk.

When we compared forecasting accuracy between models fit to one region against test data from another region, we found some interesting results. In some cases, another state's forecasted data was a better fit than the best fit model for that state. This was the case for Alaska, whose test data was more accurately forecasted by the ARIMA model fit to the Washington data. Only the Washington and Oregon model forecasts were the best fit for their state's test data. This is the opposite of what I would have expected, given our original hypothesis that Washington and Oregon's time series look very similar. I would have expected forecasts from these two states to not be accurate when tested against other states' data, since California and Alaska's data looked different, but the Washington and Oregon models performed fairly well across the board. The large regional models (PNW and total) did not perform well, which does support the idea that forecasts for salmon fisheries should be made on smaller regional scales that are ecologically relevant to salmon populations.

Description of each team member's contributions

Each member of the team took on one issue and completed the code and the analysis. Halfway through, we consulted other team members for feedback and adjust analysis based on the joint discussion. All team members participated in the lab write up and review the results.